Predictive Analytics Problem Set 4

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1. Problem to demonstrate the fitting of a binary response variable using logistic regression.

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# a) Test- Train split
library(ISLR)
## Warning: package 'ISLR' was built under R version 4.3.3
data(Weekly)

# Training: 1990-2008
train = Weekly[Weekly$Year <= 2008, ]

# Test: 2009-2010
test = Weekly[Weekly$Year > 2008, ]

train
##     Year    Lag1    Lag2    Lag3    Lag4    Lag5    Volume   Today Direction
## 1   1990   0.816   1.572  -3.936  -0.229  -3.484 0.1549760  -0.270      Down
## 2   1990  -0.270   0.816   1.572  -3.936  -0.229 0.1485740  -2.576      Down
## 3   1990  -2.576  -0.270   0.816   1.572  -3.936 0.1598375   3.514        Up
## 4   1990   3.514  -2.576  -0.270   0.816   1.572 0.1616300   0.712        Up
## 5   1990   0.712   3.514  -2.576  -0.270   0.816 0.1537280   1.178        Up
## 6   1990   1.178   0.712   3.514  -2.576  -0.270 0.1544440  -1.372      Down
## 7   1990  -1.372   1.178   0.712   3.514  -2.576 0.1517220   0.807        Up
## 8   1990   0.807  -1.372   1.178   0.712   3.514 0.1323100   0.041        Up
## 9   1990   0.041   0.807  -1.372   1.178   0.712 0.1439720   1.253        Up
## 10  1990   1.253   0.041   0.807  -1.372   1.178 0.1336350  -2.678      Down
## 11  1990  -2.678   1.253   0.041   0.807  -1.372 0.1490240  -1.793      Down
## 12  1990  -1.793  -2.678   1.253   0.041   0.807 0.1357900   2.820        Up
## 13  1990   2.820  -1.793  -2.678   1.253   0.041 0.1398980   4.022        Up
## 14  1990   4.022   2.820  -1.793  -2.678   1.253 0.1643420   0.750        Up
## 15  1990   0.750   4.022   2.820  -1.793  -2.678 0.1756480  -0.017      Down
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## 16  1990  -0.017   0.750   4.022   2.820  -1.793 0.1634700   2.420        Up
## 17  1990   2.420  -0.017   0.750   4.022   2.820 0.1726250  -1.225      Down
## 18  1990  -1.225   2.420  -0.017   0.750   4.022 0.1684460   1.171        Up
## 19  1990   1.171  -1.225   2.420  -0.017   0.750 0.1552920  -2.061      Down
## 20  1990  -2.061   1.171  -1.225   2.420  -0.017 0.1433920   0.729        Up
## 21  1990   0.729  -2.061   1.171  -1.225   2.420 0.1405540   0.112        Up
## 22  1990   0.112   0.729  -2.061   1.171  -1.225 0.1250750   2.480        Up
## 23  1990   2.480   0.112   0.729  -2.061   1.171 0.1716040  -1.552      Down
## 24  1990  -1.552   2.480   0.112   0.729  -2.061 0.1669560  -2.259      Down
## 25  1990  -2.259  -1.552   2.480   0.112   0.729 0.1717180  -2.428      Down
## 26  1990  -2.428  -2.259  -1.552   2.480   0.112 0.2098160  -2.708      Down
## 27  1990  -2.708  -2.428  -2.259  -1.552   2.480 0.1927060  -2.292      Down
## 28  1990  -2.292  -2.708  -2.428  -2.259  -1.552 0.1482520  -4.978      Down
## 29  1990  -4.978  -2.292  -2.708  -2.428  -2.259 0.1898580   3.547        Up
## 30  1990   3.547  -4.978  -2.292  -2.708  -2.428 0.1278840   0.260        Up
## 31  1990   0.260   3.547  -4.978  -2.292  -2.708 0.1157425  -2.032      Down
## 32  1990  -2.032   0.260   3.547  -4.978  -2.292 0.1239240  -1.739      Down
## 33  1990  -1.739  -2.032   0.260   3.547  -4.978 0.1490820  -1.693      Down
## 34  1990  -1.693  -1.739  -2.032   0.260   3.547 0.1718560   1.781        Up
## 35  1990   1.781  -1.693  -1.739  -2.032   0.260 0.1649700  -3.682      Down
## 36  1990  -3.682   1.781  -1.693  -1.739  -2.032 0.1564540   4.150        Up
## 37  1990   4.150  -3.682   1.781  -1.693  -1.739 0.1802800  -2.487      Down
## 38  1990  -2.487   4.150  -3.682   1.781  -1.693 0.1439780   2.343        Up
## 39  1990   2.343  -2.487   4.150  -3.682   1.781 0.1542920   0.606        Up
## 40  1990   0.606   2.343  -2.487   4.150  -3.682 0.1480060   1.077        Up
## 41  1990   1.077   0.606   2.343  -2.487   4.150 0.1635500  -0.637      Down
## 42  1990  -0.637   1.077   0.606   2.343  -2.487 0.1265325   2.260        Up
## 43  1990   2.260  -0.637   1.077   0.606   2.343 0.1515780   1.716        Up
## 44  1990   1.716   2.260  -0.637   1.077   0.606 0.1979940  -0.284      Down
## 45  1990  -0.284   1.716   2.260  -0.637   1.077 0.1558740   1.508        Up
## 46  1990   1.508  -0.284   1.716   2.260  -0.637 0.1767000  -0.913      Down
## 47  1990  -0.913   1.508  -0.284   1.716   2.260 0.0874650  -2.349      Down
## 48  1991  -2.349  -0.913   1.508  -0.284   1.716 0.1306700  -1.798      Down
## 49  1991  -1.798  -2.349  -0.913   1.508  -0.284 0.1425320   5.393        Up
## 50  1991   5.393  -1.798  -2.349  -0.913   1.508 0.1822480   1.156        Up
## 51  1991   1.156   5.393  -1.798  -2.349  -0.913 0.1798940   2.077        Up
## 52  1991   2.077   1.156   5.393  -1.798  -2.349 0.1949980   4.751        Up
## 53  1991   4.751   2.077   1.156   5.393  -1.798 0.2596560   2.702        Up
## 54  1991   2.702   4.751   2.077   1.156   5.393 0.2381400  -0.924      Down
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## 55  1991  -0.924   2.702   4.751   2.077   1.156 0.1937775   1.318        Up
## 56  1991   1.318  -0.924   2.702   4.751   2.077 0.2027840   1.209        Up
## 57  1991   1.209   1.318  -0.924   2.702   4.751 0.2239460  -0.363      Down
## 58  1991  -0.363   1.209   1.318  -0.924   2.702 0.1967520  -1.635      Down
## 59  1991  -1.635  -0.363   1.209   1.318  -0.924 0.1795400   2.106        Up
## 60  1991   2.106  -1.635  -0.363   1.209   1.318 0.1763050   0.037        Up
## 61  1991   0.037   2.106  -1.635  -0.363   1.209 0.1865580   1.343        Up
## 62  1991   1.343   0.037   2.106  -1.635  -0.363 0.1743280   0.999        Up
## 63  1991   0.999   1.343   0.037   2.106  -1.635 0.2072280  -1.348      Down
## 64  1991  -1.348   0.999   1.343   0.037   2.106 0.1641080   0.470        Up
## 65  1991   0.470  -1.348   0.999   1.343   0.037 0.1766460  -1.329      Down
## 66  1991  -1.329   0.470  -1.348   0.999   1.343 0.1585660  -0.892      Down
## 67  1991  -0.892  -1.329   0.470  -1.348   0.999 0.1718580   1.370        Up
## 68  1991   1.370  -0.892  -1.329   0.470  -1.348 0.1486320   3.269        Up
## 69  1991   3.269   1.370  -0.892  -1.329   0.470 0.2043200  -2.668      Down
## 70  1991  -2.668   3.269   1.370  -0.892  -1.329 0.1757660   0.754        Up
## 71  1991   0.754  -2.668   3.269   1.370  -0.892 0.1538140  -1.188      Down
## 72  1991  -1.188   0.754  -2.668   3.269   1.370 0.1606320  -1.745      Down
## 73  1991  -1.745  -1.188   0.754  -2.668   3.269 0.1615340   0.787        Up
## 74  1991   0.787  -1.745  -1.188   0.754  -2.668 0.1338150   1.649        Up
## 75  1991   1.649   0.787  -1.745  -1.188   0.754 0.1602280   1.044        Up
## 76  1991   1.044   1.649   0.787  -1.745  -1.188 0.1863660  -0.856      Down
## 77  1991  -0.856   1.044   1.649   0.787  -1.745 0.1483000   1.641        Up
## 78  1991   1.641  -0.856   1.044   1.649   0.787 0.1609440  -0.015      Down
## 79  1991  -0.015   1.641  -0.856   1.044   1.649 0.1564720  -0.398      Down
## 80  1991  -0.398  -0.015   1.641  -0.856   1.044 0.1693200   2.228        Up
## 81  1991   2.228  -0.398  -0.015   1.641  -0.856 0.2018520   0.320        Up
## 82  1991   0.320   2.228  -0.398  -0.015   1.641 0.1485440  -1.601      Down
## 83  1991  -1.601   0.320   2.228  -0.398  -0.015 0.1600150  -1.416      Down
## 84  1991  -1.416  -1.601   0.320   2.228  -0.398 0.1473080   1.129        Up
## 85  1991   1.129  -1.416  -1.601   0.320   2.228 0.1895540  -0.521      Down
## 86  1991  -0.521   1.129  -1.416  -1.601   0.320 0.1579680  -1.205      Down
## 87  1991  -1.205  -0.521   1.129  -1.416  -1.601 0.1630180   0.052        Up
## 88  1991   0.052  -1.205  -0.521   1.129  -1.416 0.1650700   2.897        Up
## 89  1991   2.897   0.052  -1.205  -0.521   1.129 0.1958320  -2.115      Down
## 90  1991  -2.115   2.897   0.052  -1.205  -0.521 0.1760080   1.853        Up
## 91  1991   1.853  -2.115   2.897   0.052  -1.205 0.1870600   0.401        Up
## 92  1991   0.401   1.853  -2.115   2.897   0.052 0.1767860  -2.614      Down
## 93  1991  -2.614   0.401   1.853  -2.115   2.897 0.1903460  -1.694      Down
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## 94  1991  -1.694  -2.614   0.401   1.853  -2.115 0.2123900  -0.245      Down
## 95  1991  -0.245  -1.694  -2.614   0.401   1.853 0.1585575   1.034        Up
## 96  1991   1.034  -0.245  -1.694  -2.614   0.401 0.1858220   1.417        Up
## 97  1991   1.417   1.034  -0.245  -1.694  -2.614 0.1925060   0.668        Up
## 98  1991   0.668   1.417   1.034  -0.245  -1.694 0.2144540   5.018        Up
## 99  1991   5.018   0.668   1.417   1.034  -0.245 0.1746800   3.169        Up
## 100 1992   3.169   5.018   0.668   1.417   1.034 0.2311300  -1.011      Down
## 101 1992  -1.011   3.169   5.018   0.668   1.417 0.2646440   0.906        Up
## 102 1992   0.906  -1.011   3.169   5.018   0.668 0.2809220  -0.807      Down
## 103 1992  -0.807   0.906  -1.011   3.169   5.018 0.2152000  -1.613      Down
## 104 1992  -1.613  -0.807   0.906  -1.011   3.169 0.2101220   0.565        Up
## 105 1992   0.565  -1.613  -0.807   0.906  -1.011 0.2309160   0.338        Up
## 106 1992   0.338   0.565  -1.613  -0.807   0.906 0.2133280  -0.255      Down
## 107 1992  -0.255   0.338   0.565  -1.613  -0.807 0.2498925   0.309        Up
## 108 1992   0.309  -0.255   0.338   0.565  -1.613 0.2093640  -2.001      Down
## 109 1992  -2.001   0.309  -0.255   0.338   0.565 0.1958180   0.346        Up
## 110 1992   0.346  -2.001   0.309  -0.255   0.338 0.1816380   1.345        Up
## 111 1992   1.345   0.346  -2.001   0.309  -0.255 0.1956880  -1.896      Down
## 112 1992  -1.896   1.345   0.346  -2.001   0.309 0.1768340  -0.483      Down
## 113 1992  -0.483  -1.896   1.345   0.346  -2.001 0.1753340   0.682        Up
## 114 1992   0.682  -0.483  -1.896   1.345   0.346 0.2130720   2.906        Up
## 115 1992   2.906   0.682  -0.483  -1.896   1.345 0.2093025  -1.687      Down
## 116 1992  -1.687   2.906   0.682  -0.483  -1.896 0.2120920   0.858        Up
## 117 1992   0.858  -1.687   2.906   0.682  -0.483 0.1939760   0.853        Up
## 118 1992   0.853   0.858  -1.687   2.906   0.682 0.1825480  -1.433      Down
## 119 1992  -1.433   0.853   0.858  -1.687   2.906 0.1812680   0.958        Up
## 120 1992   0.958  -1.433   0.853   0.858  -1.687 0.1736520   0.321        Up
## 121 1992   0.321   0.958  -1.433   0.853   0.858 0.1948125  -0.450      Down
## 122 1992  -0.450   0.321   0.958  -1.433   0.853 0.2005260  -0.900      Down
## 123 1992  -0.900  -0.450   0.321   0.958  -1.433 0.1899600  -1.486      Down
## 124 1992  -1.486  -0.900  -0.450   0.321   0.958 0.2090600  -0.054      Down
## 125 1992  -0.054  -1.486  -0.900  -0.450   0.321 0.1779640   2.062        Up
## 126 1992   2.062  -0.054  -1.486  -0.900  -0.450 0.2016825   0.692        Up
## 127 1992   0.692   2.062  -0.054  -1.486  -0.900 0.1973500   0.241        Up
## 128 1992   0.241   0.692   2.062  -0.054  -1.486 0.1900040  -0.967      Down
## 129 1992  -0.967   0.241   0.692   2.062  -0.054 0.1738120   3.064        Up
## 130 1992   3.064  -0.967   0.241   0.692   2.062 0.2049880  -1.256      Down
## 131 1992  -1.256   3.064  -0.967   0.241   0.692 0.1751500   0.246        Up
## 132 1992   0.246  -1.256   3.064  -0.967   0.241 0.1691100  -1.205      Down
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## 133 1992  -1.205   0.246  -1.256   3.064  -0.967 0.1799740  -0.002      Down
## 134 1992  -0.002  -1.205   0.246  -1.256   3.064 0.1742340   0.540        Up
## 135 1992   0.540  -0.002  -1.205   0.246  -1.256 0.1717040   0.599        Up
## 136 1992   0.599   0.540  -0.002  -1.205   0.246 0.1856975   0.798        Up
## 137 1992   0.798   0.599   0.540  -0.002  -1.205 0.2239920  -2.029      Down
## 138 1992  -2.029   0.798   0.599   0.540  -0.002 0.1900160  -0.936      Down
## 139 1992  -0.936  -2.029   0.798   0.599   0.540 0.1813580  -1.903      Down
## 140 1992  -1.903  -0.936  -2.029   0.798   0.599 0.2116740   2.253        Up
## 141 1992   2.253  -1.903  -0.936  -2.029   0.798 0.1877460   0.576        Up
## 142 1992   0.576   2.253  -1.903  -0.936  -2.029 0.2229840   1.106        Up
## 143 1992   1.106   0.576   2.253  -1.903  -0.936 0.2004360  -0.263      Down
## 144 1992  -0.263   1.106   0.576   2.253  -1.903 0.2061720   1.161        Up
## 145 1992   1.161  -0.263   1.106   0.576   2.253 0.2166900   0.999        Up
## 146 1992   0.999   1.161  -0.263   1.106   0.576 0.2113040   0.823        Up
## 147 1992   0.823   0.999   1.161  -0.263   1.106 0.1869475   0.442        Up
## 148 1992   0.442   0.823   0.999   1.161  -0.263 0.2418440   0.387        Up
## 149 1992   0.387   0.442   0.823   0.999   1.161 0.2174480   1.741        Up
## 150 1992   1.741   0.387   0.442   0.823   0.999 0.2595760  -0.342      Down
## 151 1992  -0.342   1.741   0.387   0.442   0.823 0.2011225  -0.923      Down
## 152 1993  -0.923  -0.342   1.741   0.387   0.442 0.1768675  -1.529      Down
## 153 1993  -1.529  -0.923  -0.342   1.741   0.387 0.2610240   1.888        Up
## 154 1993   1.888  -1.529  -0.923  -0.342   1.741 0.2585360  -0.238      Down
## 155 1993  -0.238   1.888  -1.529  -0.923  -0.342 0.2598000   0.612        Up
## 156 1993   0.612  -0.238   1.888  -1.529  -0.923 0.2768100   2.313        Up
## 157 1993   2.313   0.612  -0.238   1.888  -1.529 0.3062780  -0.969      Down
## 158 1993  -0.969   2.313   0.612  -0.238   1.888 0.2419440  -2.330      Down
## 159 1993  -2.330  -0.969   2.313   0.612  -0.238 0.3142350   2.110        Up
## 160 1993   2.110  -2.330  -0.969   2.313   0.612 0.2888800   0.616        Up
## 161 1993   0.616   2.110  -2.330  -0.969   2.313 0.2534580   0.834        Up
## 162 1993   0.834   0.616   2.110  -2.330  -0.969 0.2668100   0.078        Up
## 163 1993   0.078   0.834   0.616   2.110  -2.330 0.2473720  -0.533      Down
## 164 1993  -0.533   0.078   0.834   0.616   2.110 0.2436800  -1.427      Down
## 165 1993  -1.427  -0.533   0.078   0.834   0.616 0.2536420   0.102        Up
## 166 1993   0.102  -1.427  -0.533   0.078   0.834 0.2935325   1.607        Up
## 167 1993   1.607   0.102  -1.427  -0.533   0.078 0.2736760  -2.653      Down
## 168 1993  -2.653   1.607   0.102  -1.427  -0.533 0.2840400   0.723        Up
## 169 1993   0.723  -2.653   1.607   0.102  -1.427 0.2665200   0.482        Up
## 170 1993   0.482   0.723  -2.653   1.607   0.102 0.2493100  -0.622      Down
## 171 1993  -0.622   0.482   0.723  -2.653   1.607 0.2513140   1.429        Up
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## 172 1993   1.429  -0.622   0.482   0.723  -2.653 0.2805160   0.976        Up
## 173 1993   0.976   1.429  -0.622   0.482   0.723 0.2405880  -0.029      Down
## 174 1993  -0.029   0.976   1.429  -0.622   0.482 0.2593150  -0.622      Down
## 175 1993  -0.622  -0.029   0.976   1.429  -0.622 0.2431880  -0.800      Down
## 176 1993  -0.800  -0.622  -0.029   0.976   1.429 0.2504720   0.884        Up
## 177 1993   0.884  -0.800  -0.622  -0.029   0.976 0.2478640  -0.393      Down
## 178 1993  -0.393   0.884  -0.800  -0.622  -0.029 0.2624620   0.509        Up
## 179 1993   0.509  -0.393   0.884  -0.800  -0.622 0.2515250  -0.527      Down
## 180 1993  -0.527   0.509  -0.393   0.884  -0.800 0.2554740   0.303        Up
## 181 1993   0.303  -0.527   0.509  -0.393   0.884 0.2488360   0.230        Up
## 182 1993   0.230   0.303  -0.527   0.509  -0.393 0.2536180   0.123        Up
## 183 1993   0.123   0.230   0.303  -0.527   0.509 0.2393200   0.325        Up
## 184 1993   0.325   0.123   0.230   0.303  -0.527 0.2499000   1.337        Up
## 185 1993   1.337   0.325   0.123   0.230   0.303 0.2756060   0.960        Up
## 186 1993   0.960   1.337   0.325   0.123   0.230 0.2470120   0.174        Up
## 187 1993   0.174   0.960   1.337   0.325   0.123 0.2298160   0.082        Up
## 188 1993   0.082   0.174   0.960   1.337   0.325 0.2601550  -0.626      Down
## 189 1993  -0.626   0.082   0.174   0.960   1.337 0.2818200  -0.262      Down
## 190 1993  -0.262  -0.626   0.082   0.174   0.960 0.2708040   0.798        Up
## 191 1993   0.798  -0.262  -0.626   0.082   0.174 0.2607580  -0.210      Down
## 192 1993  -0.210   0.798  -0.262  -0.626   0.082 0.2599660   1.996        Up
## 193 1993   1.996  -0.210   0.798  -0.262  -0.626 0.2913200  -1.327      Down
## 194 1993  -1.327   1.996  -0.210   0.798  -0.262 0.3061380   0.984        Up
## 195 1993   0.984  -1.327   1.996  -0.210   0.798 0.2792920  -1.766      Down
## 196 1993  -1.766   0.984  -1.327   1.996  -0.210 0.3126480   1.266        Up
## 197 1993   1.266  -1.766   0.984  -1.327   1.996 0.2808420  -0.599      Down
## 198 1993  -0.599   1.266  -1.766   0.984  -1.327 0.2976820   0.099        Up
## 199 1993   0.099  -0.599   1.266  -1.766   0.984 0.2153450   0.395        Up
## 200 1993   0.395   0.099  -0.599   1.266  -1.766 0.2755940  -0.207      Down
## 201 1993  -0.207   0.395   0.099  -0.599   1.266 0.2851420   0.528        Up
## 202 1993   0.528  -0.207   0.395   0.099  -0.599 0.3023540   0.214        Up
## 203 1993   0.214   0.528  -0.207   0.395   0.099 0.2572375  -0.199      Down
## 204 1994  -0.199   0.214   0.528  -0.207   0.395 0.2012360   0.740        Up
## 205 1994   0.740  -0.199   0.214   0.528  -0.207 0.3375300   1.066        Up
## 206 1994   1.066   0.740  -0.199   0.214   0.528 0.3037120  -0.040      Down
## 207 1994  -0.040   1.066   0.740  -0.199   0.214 0.3021980   0.838        Up
## 208 1994   0.838  -0.040   1.066   0.740  -0.199 0.3174640  -1.857      Down
## 209 1994  -1.857   0.838  -0.040   1.066   0.740 0.3342140   0.079        Up
## 210 1994   0.079  -1.857   0.838  -0.040   1.066 0.3080220  -0.530      Down
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## 211 1994  -0.530   0.079  -1.857   0.838  -0.040 0.2997340  -0.346       Down
## 212 1994  -0.346  -0.530   0.079  -1.857   0.838 0.2993575  -0.285       Down
## 213 1994  -0.285  -0.346  -0.530   0.079  -1.857 0.3075820   0.366         Up
## 214 1994   0.366  -0.285  -0.346  -0.530   0.079 0.3133540   0.990         Up
## 215 1994   0.990   0.366  -0.285  -0.346  -0.530 0.3275420  -2.225       Down
## 216 1994  -2.225   0.990   0.366  -0.285  -0.346 0.2729000  -3.216       Down
## 217 1994  -3.216  -2.225   0.990   0.366  -0.285 0.3467025   0.298         Up
## 218 1994   0.298  -3.216  -2.225   0.990   0.366 0.3131500  -0.206       Down
## 219 1994  -0.206   0.298  -3.216  -2.225   0.990 0.2727760   0.325         Up
## 220 1994   0.325  -0.206   0.298  -3.216  -2.225 0.3271540   0.733         Up
## 221 1994   0.733   0.325  -0.206   0.298  -3.216 0.2924025  -0.685       Down
## 222 1994  -0.685   0.733   0.325  -0.206   0.298 0.2799880  -0.822       Down
## 223 1994  -0.822  -0.685   0.733   0.325  -0.206 0.2701540   2.427         Up
## 224 1994   2.427  -0.822  -0.685   0.733   0.325 0.2965020   0.530         Up
## 225 1994   0.530   2.427  -0.822  -0.685   0.733 0.2452100   0.612         Up
## 226 1994   0.612   0.530   2.427  -0.822  -0.685 0.2599325  -0.317       Down
## 227 1994  -0.317   0.612   0.530   2.427  -0.822 0.2450220  -0.048       Down
## 228 1994  -0.048  -0.317   0.612   0.530   2.427 0.2863540  -3.414       Down
## 229 1994  -3.414  -0.048  -0.317   0.612   0.530 0.2594200   0.768         Up
## 230 1994   0.768  -3.414  -0.048  -0.317   0.612 0.2549380   0.751         Up
## 231 1994   0.751   0.768  -3.414  -0.048  -0.317 0.2319750   1.025         Up
## 232 1994   1.025   0.751   0.768  -3.414  -0.048 0.2678500  -0.231       Down
## 233 1994  -0.231   1.025   0.751   0.768  -3.414 0.2601100   1.137         Up
## 234 1994   1.137  -0.231   1.025   0.751   0.768 0.2426740  -0.255       Down
## 235 1994  -0.255   1.137  -0.231   1.025   0.751 0.2712360   1.061         Up
## 236 1994   1.061  -0.255   1.137  -0.231   1.025 0.2562580   0.377         Up
## 237 1994   0.377   1.061  -0.255   1.137  -0.231 0.2806120   2.183         Up
## 238 1994   2.183   0.377   1.061  -0.255   1.137 0.2885940  -0.593       Down
## 239 1994  -0.593   2.183   0.377   1.061  -0.255 0.2828460  -0.597       Down
## 240 1994  -0.597  -0.593   2.183   0.377   1.061 0.2695925   0.643         Up
## 241 1994   0.643  -0.597  -0.593   2.183   0.377 0.3056400  -2.445       Down
## 242 1994  -2.445   0.643  -0.597  -0.593   2.183 0.3120520   0.661         Up
## 243 1994   0.661  -2.445   0.643  -0.597  -0.593 0.2974120  -1.645       Down
## 244 1994  -1.645   0.661  -2.445   0.643  -0.597 0.3022540   3.076         Up
## 245 1994   3.076  -1.645   0.661  -2.445   0.643 0.2855740  -0.897       Down
## 246 1994  -0.897   3.076  -1.645   0.661  -2.445 0.2924040   1.910         Up
## 247 1994   1.910  -0.897   3.076  -1.645   0.661 0.3281440  -2.425       Down
## 248 1994  -2.425   1.910  -0.897   3.076  -1.645 0.3029700   0.015         Up
## 249 1994   0.015  -2.425   1.910  -0.897   3.076 0.2770760  -0.190       Down
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## 250 1994  -0.190   0.015  -2.425   1.910  -0.897 0.3147460  -1.989      Down
## 251 1994  -1.989  -0.190   0.015  -2.425   1.910 0.3073375   0.223        Up
## 252 1994   0.223  -1.989  -0.190   0.015  -2.425 0.2842840  -1.399      Down
## 253 1994  -1.399   0.223  -1.989  -0.190   0.015 0.3079280   2.649        Up
## 254 1994   2.649  -1.399   0.223  -1.989  -0.190 0.3524980   0.224        Up
## 255 1994   0.224   2.649  -1.399   0.223  -1.989 0.3028760  -0.122      Down
## 256 1995  -0.122   0.224   2.649  -1.399   0.223 0.2410875   0.307        Up
## 257 1995   0.307  -0.122   0.224   2.649  -1.399 0.2997700   1.148        Up
## 258 1995   1.148   0.307  -0.122   0.224   2.649 0.3254660  -0.255      Down
## 259 1995  -0.255   1.148   0.307  -0.122   0.224 0.3334800   1.207        Up
## 260 1995   1.207  -0.255   1.148   0.307  -0.122 0.3256220   1.756        Up
## 261 1995   1.756   1.207  -0.255   1.148   0.307 0.3777120   0.587        Up
## 262 1995   0.587   1.756   1.207  -0.255   1.148 0.3159840   0.106        Up
## 263 1995   0.106   0.587   1.756   1.207  -0.255 0.3287980   1.274        Up
## 264 1995   1.274   0.106   0.587   1.756   1.207 0.3361900  -0.551      Down
## 265 1995  -0.551   1.274   0.106   0.587   1.756 0.3252960   0.855        Up
## 266 1995   0.855  -0.551   1.274   0.106   0.587 0.3412920   1.215        Up
## 267 1995   1.215   0.855  -0.551   1.274   0.106 0.3370060   1.100        Up
## 268 1995   1.100   1.215   0.855  -0.551   1.274 0.3317740  -0.052      Down
## 269 1995  -0.052   1.100   1.215   0.855  -0.551 0.3437140   1.140        Up
## 270 1995   1.140  -0.052   1.100   1.215   0.855 0.3154800   0.555        Up
## 271 1995   0.555   1.140  -0.052   1.100   1.215 0.3002750  -0.145      Down
## 272 1995  -0.145   0.555   1.140  -0.052   1.100 0.3656720   1.223        Up
## 273 1995   1.223  -0.145   0.555   1.140  -0.052 0.3400340   1.051        Up
## 274 1995   1.051   1.223  -0.145   0.555   1.140 0.3538260   1.044        Up
## 275 1995   1.044   1.051   1.223  -0.145   0.555 0.3472000  -1.210      Down
## 276 1995  -1.210   1.044   1.051   1.223  -0.145 0.3472520   0.859        Up
## 277 1995   0.859  -1.210   1.044   1.051   1.223 0.3346200   1.692        Up
## 278 1995   1.692   0.859  -1.210   1.044   1.051 0.3382800  -0.858      Down
## 279 1995  -0.858   1.692   0.859  -1.210   1.044 0.3246500   2.252        Up
## 280 1995   2.252  -0.858   1.692   0.859  -1.210 0.3475580   1.830        Up
## 281 1995   1.830   2.252  -0.858   1.692   0.859 0.3692460  -0.902      Down
## 282 1995  -0.902   1.830   2.252  -0.858   1.692 0.3272920   2.133        Up
## 283 1995   2.133  -0.902   1.830   2.252  -0.858 0.3406975   0.633        Up
## 284 1995   0.633   2.133  -0.902   1.830   2.252 0.3806520  -1.120      Down
## 285 1995  -1.120   0.633   2.133  -0.902   1.830 0.3999660   1.682        Up
## 286 1995   1.682  -1.120   0.633   2.133  -0.902 0.3500260  -0.709      Down
## 287 1995  -0.709   1.682  -1.120   0.633   2.133 0.3332680  -0.685      Down
## 288 1995  -0.685  -0.709   1.682  -1.120   0.633 0.2922080   0.739        Up
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## 289 1995   0.739  -0.685  -0.709   1.682  -1.120 0.3320220   0.159        Up
## 290 1995   0.159   0.739  -0.685  -0.709   1.682 0.2882340   0.668        Up
## 291 1995   0.668   0.159   0.739  -0.685  -0.709 0.2933280   1.568        Up
## 292 1995   1.568   0.668   0.159   0.739  -0.685 0.3354675   1.863        Up
## 293 1995   1.863   1.568   0.668   0.159   0.739 0.3736020  -0.278      Down
## 294 1995  -0.278   1.863   1.568   0.668   0.159 0.3670400   0.461        Up
## 295 1995   0.461  -0.278   1.863   1.568   0.668 0.3502040  -0.329      Down
## 296 1995  -0.329   0.461  -0.278   1.863   1.568 0.3422940   0.345        Up
## 297 1995   0.345  -0.329   0.461  -0.278   1.863 0.3495020   0.506        Up
## 298 1995   0.506   0.345  -0.329   0.461  -0.278 0.3728760  -1.321      Down
## 299 1995  -1.321   0.506   0.345  -0.329   0.461 0.4046820   1.875        Up
## 300 1995   1.875  -1.321   0.506   0.345  -0.329 0.3640420   0.364        Up
## 301 1995   0.364   1.875  -1.321   0.506   0.345 0.3426020   1.240        Up
## 302 1995   1.240   0.364   1.875  -1.321   0.506 0.3773680  -0.017      Down
## 303 1995  -0.017   1.240   0.364   1.875  -1.321 0.3180800   1.168        Up
## 304 1995   1.168  -0.017   1.240   0.364   1.875 0.3999260   1.730        Up
## 305 1995   1.730   1.168  -0.017   1.240   0.364 0.3935560  -0.185      Down
## 306 1995  -0.185   1.730   1.168  -0.017   1.240 0.4418640  -0.712      Down
## 307 1995  -0.712  -0.185   1.730   1.168  -0.017 0.4095280   0.650        Up
## 308 1996   0.650  -0.712  -0.185   1.730   1.168 0.2698725   0.127        Up
## 309 1996   0.127   0.650  -0.712  -0.185   1.730 0.4457050  -2.416      Down
## 310 1996  -2.416   0.127   0.650  -0.712  -0.185 0.3673580   1.665        Up
## 311 1996   1.665  -2.416   0.127   0.650  -0.712 0.4276500   1.600        Up
## 312 1996   1.600   1.665  -2.416   0.127   0.650 0.4260600   2.288        Up
## 313 1996   2.288   1.600   1.665  -2.416   0.127 0.4362680   3.229        Up
## 314 1996   3.229   2.288   1.600   1.665  -2.416 0.4518080  -1.278      Down
## 315 1996  -1.278   3.229   2.288   1.600   1.665 0.4244220   1.713        Up
## 316 1996   1.713  -1.278   3.229   2.288   1.600 0.4389325  -2.232      Down
## 317 1996  -2.232   1.713  -1.278   3.229   2.288 0.4406220  -1.687      Down
## 318 1996  -1.687  -2.232   1.713  -1.278   3.229 0.4527060   1.252        Up
## 319 1996   1.252  -1.687  -2.232   1.713  -1.278 0.4680220   1.433        Up
## 320 1996   1.433   1.252  -1.687  -2.232   1.713 0.3963500  -0.787      Down
## 321 1996  -0.787   1.433   1.252  -1.687  -2.232 0.3854660   1.605        Up
## 322 1996   1.605  -0.787   1.433   1.252  -1.687 0.3921950  -2.920      Down
## 323 1996  -2.920   1.605  -0.787   1.433   1.252 0.4493460   1.313        Up
## 324 1996   1.313  -2.920   1.605  -0.787   1.433 0.4231440   1.301        Up
## 325 1996   1.301   1.313  -2.920   1.605  -0.787 0.4413860  -1.810      Down
## 326 1996  -1.810   1.301   1.313  -2.920   1.605 0.4038020   1.630        Up
## 327 1996   1.630  -1.810   1.301   1.313  -2.920 0.4229460   2.579        Up
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## 328 1996   2.579   1.630  -1.810   1.301   1.313 0.4247240   1.435        Up
## 329 1996   1.435   2.579   1.630  -1.810   1.301 0.3958560  -1.384      Down
## 330 1996  -1.384   1.435   2.579   1.630  -1.810 0.3554800   0.626        Up
## 331 1996   0.626  -1.384   1.435   2.579   1.630 0.3995040  -1.108      Down
## 332 1996  -1.108   0.626  -1.384   1.435   2.579 0.3856620   0.149        Up
## 333 1996   0.149  -1.108   0.626  -1.384   1.435 0.4033420   0.568        Up
## 334 1996   0.568   0.149  -1.108   0.626  -1.384 0.3976600  -1.967      Down
## 335 1996  -1.967   0.568   0.149  -1.108   0.626 0.3128700  -1.711      Down
## 336 1996  -1.711  -1.967   0.568   0.149  -1.108 0.4212580  -1.154      Down
## 337 1996  -1.154  -1.711  -1.967   0.568   0.149 0.4996720  -0.443      Down
## 338 1996  -0.443  -1.154  -1.711  -1.967   0.568 0.3935040   4.181        Up
## 339 1996   4.181  -0.443  -1.154  -1.711  -1.967 0.3814800  -0.059      Down
## 340 1996  -0.059   4.181  -0.443  -1.154  -1.711 0.3421440   0.470        Up
## 341 1996   0.470  -0.059   4.181  -0.443  -1.154 0.3359400   0.274        Up
## 342 1996   0.274   0.470  -0.059   4.181  -0.443 0.3281640  -2.255      Down
## 343 1996  -2.255   0.274   0.470  -0.059   4.181 0.2935780   0.566        Up
## 344 1996   0.566  -2.255   0.274   0.470  -0.059 0.3517925   3.791        Up
## 345 1996   3.791   0.566  -2.255   0.274   0.470 0.3897100   0.954        Up
## 346 1996   0.954   3.791   0.566  -2.255   0.274 0.4389060  -0.122      Down
## 347 1996  -0.122   0.954   3.791   0.566  -2.255 0.4250500   2.225        Up
## 348 1996   2.225  -0.122   0.954   3.791   0.566 0.4201380  -0.114      Down
## 349 1996  -0.114   2.225  -0.122   0.954   3.791 0.4030540   1.450        Up
## 350 1996   1.450  -0.114   2.225  -0.122   0.954 0.4347920  -1.393      Down
## 351 1996  -1.393   1.450  -0.114   2.225  -0.122 0.4108400   0.407        Up
## 352 1996   0.407  -1.393   1.450  -0.114   2.225 0.4427260   3.844        Up
## 353 1996   3.844   0.407  -1.393   1.450  -0.114 0.4599800   0.930        Up
## 354 1996   0.930   3.844   0.407  -1.393   1.450 0.4529980   1.506        Up
## 355 1996   1.506   0.930   3.844   0.407  -1.393 0.4676080   1.107        Up
## 356 1996   1.107   1.506   0.930   3.844   0.407 0.3488525  -2.301      Down
## 357 1996  -2.301   1.107   1.506   0.930   3.844 0.4822980  -1.482      Down
## 358 1996  -1.482  -2.301   1.107   1.506   0.930 0.4546720   2.776        Up
## 359 1996   2.776  -1.482  -2.301   1.107   1.506 0.5297280   1.058        Up
## 360 1996   1.058   2.776  -1.482  -2.301   1.107 0.2542150  -1.158      Down
## 361 1997  -1.158   1.058   2.776  -1.482  -2.301 0.4137550   1.533        Up
## 362 1997   1.533  -1.158   1.058   2.776  -1.482 0.5456600   2.195        Up
## 363 1997   2.195   1.533  -1.158   1.058   2.776 0.5147840  -0.728      Down
## 364 1997  -0.728   2.195   1.533  -1.158   1.058 0.5657940   2.030        Up
## 365 1997   2.030  -0.728   2.195   1.533  -1.158 0.5176880   0.432        Up
## 366 1997   0.432   2.030  -0.728   2.195   1.533 0.5222440   2.396        Up
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## 367 1997   2.396   0.432   2.030  -0.728   2.195 0.5207640  -0.830      Down
## 368 1997  -0.830   2.396   0.432   2.030  -0.728 0.4910325  -1.366      Down
## 369 1997  -1.366  -0.830   2.396   0.432   2.030 0.5073520   1.789        Up
## 370 1997   1.789  -1.366  -0.830   2.396   0.432 0.5112380  -1.466      Down
## 371 1997  -1.466   1.789  -1.366  -0.830   2.396 0.4902660  -1.144      Down
## 372 1997  -1.144  -1.466   1.789  -1.366  -0.830 0.5268820  -1.303      Down
## 373 1997  -1.303  -1.144  -1.466   1.789  -1.366 0.4807375  -2.065      Down
## 374 1997  -2.065  -1.303  -1.144  -1.466   1.789 0.5184900  -2.672      Down
## 375 1997  -2.672  -2.065  -1.303  -1.144  -1.466 0.4444500   3.889        Up
## 376 1997   3.889  -2.672  -2.065  -1.303  -1.144 0.4771380  -0.127      Down
## 377 1997  -0.127   3.889  -2.672  -2.065  -1.303 0.4604280   6.219        Up
## 378 1997   6.219  -0.127   3.889  -2.672  -2.065 0.4936760   1.453        Up
## 379 1997   1.453   6.219  -0.127   3.889  -2.672 0.5286960   0.603        Up
## 380 1997   0.603   1.453   6.219  -0.127   3.889 0.4798080   2.083        Up
## 381 1997   2.083   0.603   1.453   6.219  -0.127 0.4361380   0.148        Up
## 382 1997   0.148   2.083   0.603   1.453   6.219 0.4808225   1.147        Up
## 383 1997   1.147   0.148   2.083   0.603   1.453 0.4742620   4.110        Up
## 384 1997   4.110   1.147   0.148   2.083   0.603 0.5350140   0.608        Up
## 385 1997   0.608   4.110   1.147   0.148   2.083 0.5278160  -1.268      Down
## 386 1997  -1.268   0.608   4.110   1.147   0.148 0.5221900   3.338        Up
## 387 1997   3.338  -1.268   0.608   4.110   1.147 0.5018450  -0.026      Down
## 388 1997  -0.026   3.338  -1.268   0.608   4.110 0.5370580  -0.151      Down
## 389 1997  -0.151  -0.026   3.338  -1.268   0.608 0.5901360   2.566        Up
## 390 1997   2.566  -0.151  -0.026   3.338  -1.268 0.5497100   0.889        Up
## 391 1997   0.889   2.566  -0.151  -0.026   3.338 0.5283020  -1.436      Down
## 392 1997  -1.436   0.889   2.566  -0.151  -0.026 0.5372720  -3.506      Down
## 393 1997  -3.506  -1.436   0.889   2.566  -0.151 0.5270280   2.523        Up
## 394 1997   2.523  -3.506  -1.436   0.889   2.566 0.5080780  -2.606      Down
## 395 1997  -2.606   2.523  -3.506  -1.436   0.889 0.4460920   3.289        Up
## 396 1997   3.289  -2.606   2.523  -3.506  -1.436 0.5341600  -0.553      Down
## 397 1997  -0.553   3.289  -2.606   2.523  -3.506 0.5210840   2.879        Up
## 398 1997   2.879  -0.553   3.289  -2.606   2.523 0.5785660  -0.557      Down
## 399 1997  -0.557   2.879  -0.553   3.289  -2.606 0.5367020   2.096        Up
## 400 1997   2.096  -0.557   2.879  -0.553   3.289 0.5522780   0.202        Up
## 401 1997   0.202   2.096  -0.557   2.879  -0.553 0.5346440  -2.360      Down
## 402 1997  -2.360   0.202   2.096  -0.557   2.879 0.5184860  -0.267      Down
## 403 1997  -0.267  -2.360   0.202   2.096  -0.557 0.6061160  -2.869      Down
## 404 1997  -2.869  -0.267  -2.360   0.202   2.096 0.8048480   1.409        Up
## 405 1997   1.409  -2.869  -0.267  -2.360   0.202 0.5529760   0.091        Up
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## 406 1997   0.091   1.409  -2.869  -0.267  -2.360 0.5549720   3.742        Up
## 407 1997   3.742   0.091   1.409  -2.869  -0.267 0.5708500  -0.798      Down
## 408 1997  -0.798   3.742   0.091   1.409  -2.869 0.4449075   2.972        Up
## 409 1997   2.972  -0.798   3.742   0.091   1.409 0.5976180  -3.090      Down
## 410 1997  -3.090   2.972  -0.798   3.742   0.091 0.5685580  -0.693      Down
## 411 1997  -0.693  -3.090   2.972  -0.798   3.742 0.6502880  -1.090      Down
## 412 1997  -1.090  -0.693  -3.090   2.972  -0.798 0.3666550   4.120        Up
## 413 1998   4.120  -1.090  -0.693  -3.090   2.972 0.4441675  -4.856      Down
## 414 1998  -4.856   4.120  -1.090  -0.693  -3.090 0.6624760   3.646        Up
## 415 1998   3.646  -4.856   4.120  -1.090  -0.693 0.6389200  -0.408      Down
## 416 1998  -0.408   3.646  -4.856   4.120  -1.090 0.6383225   2.369        Up
## 417 1998   2.369  -0.408   3.646  -4.856   4.120 0.6613660   3.283        Up
## 418 1998   3.283   2.369  -0.408   3.646  -4.856 0.6770980   0.754        Up
## 419 1998   0.754   3.283   2.369  -0.408   3.646 0.5820660   1.384        Up
## 420 1998   1.384   0.754   3.283   2.369  -0.408 0.5970025   1.463        Up
## 421 1998   1.463   1.384   0.754   3.283   2.369 0.5945440   0.605        Up
## 422 1998   0.605   1.463   1.384   0.754   3.283 0.6323760   1.224        Up
## 423 1998   1.224   0.605   1.463   1.384   0.754 0.6209240   2.859        Up
## 424 1998   2.859   1.224   0.605   1.463   1.384 0.6356360  -0.338      Down
## 425 1998  -0.338   2.859   1.224   0.605   1.463 0.6205160   2.488        Up
## 426 1998   2.488  -0.338   2.859   1.224   0.605 0.6355720  -1.072      Down
## 427 1998  -1.072   2.488  -0.338   2.859   1.224 0.6154600   1.085        Up
## 428 1998   1.085  -1.072   2.488  -0.338   2.859 0.6470180  -1.320      Down
## 429 1998  -1.320   1.085  -1.072   2.488  -0.338 0.6509300   1.182        Up
## 430 1998   1.182  -1.320   1.085  -1.072   2.488 0.6561840  -1.147      Down
## 431 1998  -1.147   1.182  -1.320   1.085  -1.072 0.5784000   0.053        Up
## 432 1998   0.053  -1.147   1.182  -1.320   1.085 0.5931280   0.157        Up
## 433 1998   0.157   0.053  -1.147   1.182  -1.320 0.5336800  -1.770      Down
## 434 1998  -1.770   0.157   0.053  -1.147   1.182 0.5922825   2.112        Up
## 435 1998   2.112  -1.770   0.157   0.053  -1.147 0.5697960  -1.348      Down
## 436 1998  -1.348   2.112  -1.770   0.157   0.053 0.5954360   0.165        Up
## 437 1998   0.165  -1.348   2.112  -1.770   0.157 0.6621520   2.957        Up
## 438 1998   2.957   0.165  -1.348   2.112  -1.770 0.6187000   1.167        Up
## 439 1998   1.167   2.957   0.165  -1.348   2.112 0.6333400   1.562        Up
## 440 1998   1.562   1.167   2.957   0.165  -1.348 0.5973100   1.926        Up
## 441 1998   1.926   1.562   1.167   2.957   0.165 0.6589820  -3.872      Down
## 442 1998  -3.872   1.926   1.562   1.167   2.957 0.6800560  -1.765      Down
## 443 1998  -1.765  -3.872   1.926   1.562   1.167 0.6602500  -2.786      Down
## 444 1998  -2.786  -1.765  -3.872   1.926   1.562 0.7704200  -2.451      Down
```

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## 445 1998  -2.451  -2.786  -1.765  -3.872   1.926 0.6740020   1.740        Up
## 446 1998   1.740  -2.451  -2.786  -1.765  -3.872 0.6511880  -5.004      Down
## 447 1998  -5.004   1.740  -2.451  -2.786  -1.765 0.7352000  -5.184      Down
## 448 1998  -5.184  -5.004   1.740  -2.451  -2.786 0.9379000   3.611        Up
## 449 1998   3.611  -5.184  -5.004   1.740  -2.451 0.8046250   1.093        Up
## 450 1998   1.093   3.611  -5.184  -5.004   1.740 0.7451400   2.417        Up
## 451 1998   2.417   1.093   3.611  -5.184  -5.004 0.7494360  -4.034      Down
## 452 1998  -4.034   2.417   1.093   3.611  -5.184 0.8106600  -1.816      Down
## 453 1998  -1.816  -4.034   2.417   1.093   3.611 0.9265800   7.317        Up
## 454 1998   7.317  -1.816  -4.034   2.417   1.093 0.8390800   1.349        Up
## 455 1998   1.349   7.317  -1.816  -4.034   2.417 0.7668880   2.615        Up
## 456 1998   2.615   1.349   7.317  -1.816  -4.034 0.7072620   3.854        Up
## 457 1998   3.854   2.615   1.349   7.317  -1.816 0.7545000  -1.340      Down
## 458 1998  -1.340   3.854   2.615   1.349   7.317 0.6489120   3.361        Up
## 459 1998   3.361  -1.340   3.854   2.615   1.349 0.6730980   2.473        Up
## 460 1998   2.473   3.361  -1.340   3.854   2.615 0.5952075  -1.308      Down
## 461 1998  -1.308   2.473   3.361  -1.340   3.854 0.7426600  -0.874      Down
## 462 1998  -0.874  -1.308   2.473   3.361  -1.340 0.7061200   1.849        Up
## 463 1998   1.849  -0.874  -1.308   2.473   3.361 0.7648400   3.219        Up
## 464 1998   3.219   1.849  -0.874  -1.308   2.473 0.5924450   0.241        Up
## 465 1999   0.241   3.219   1.849  -0.874  -1.308 0.6078675   3.731        Up
## 466 1999   3.731   0.241   3.219   1.849  -0.874 0.8879400  -2.496      Down
## 467 1999  -2.496   3.731   0.241   3.219   1.849 0.8290000  -1.453      Down
## 468 1999  -1.453  -2.496   3.731   0.241   3.219 0.8372250   4.444        Up
## 469 1999   4.444  -1.453  -2.496   3.731   0.241 0.8559800  -3.145      Down
## 470 1999  -3.145   4.444  -1.453  -2.496   3.731 0.8495600  -0.748      Down
## 471 1999  -0.748  -3.145   4.444  -1.453  -2.496 0.7340200   0.739        Up
## 472 1999   0.739  -0.748  -3.145   4.444  -1.453 0.7078150  -0.072      Down
## 473 1999  -0.072   0.739  -0.748  -3.145   4.444 0.7613400   2.999        Up
## 474 1999   2.999  -0.072   0.739  -0.748  -3.145 0.7621200   1.499        Up
## 475 1999   1.499   2.999  -0.072   0.739  -0.748 0.8181600   0.363        Up
## 476 1999   0.363   1.499   2.999  -0.072   0.739 0.7954200  -1.269      Down
## 477 1999  -1.269   0.363   1.499   2.999  -0.072 0.7445600   0.851        Up
## 478 1999   0.851  -1.269   0.363   1.499   2.999 0.7760500   4.223        Up
## 479 1999   4.223   0.851  -1.269   0.363   1.499 0.7732600  -2.177      Down
## 480 1999  -2.177   4.223   0.851  -1.269   0.363 0.9331600   2.870        Up
## 481 1999   2.870  -2.177   4.223   0.851  -1.269 0.9585200  -1.597      Down
## 482 1999  -1.597   2.870  -2.177   4.223   0.851 0.8991000   0.735        Up
## 483 1999   0.735  -1.597   2.870  -2.177   4.223 0.8696600  -0.535      Down
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## 484 1999  -0.535   0.735  -1.597   2.870  -2.177 0.7919200  -0.561      Down
## 485 1999  -0.561  -0.535   0.735  -1.597   2.870 0.7317600  -2.139      Down
## 486 1999  -2.139  -0.561  -0.535   0.735  -1.597 0.7827120   1.990        Up
## 487 1999   1.990  -2.139  -0.561  -0.535   0.735 0.7064750  -2.569      Down
## 488 1999  -2.569   1.990  -2.139  -0.561  -0.535 0.6853800   3.803        Up
## 489 1999   3.803  -2.569   1.990  -2.139  -0.561 0.7575200  -2.050      Down
## 490 1999  -2.050   3.803  -2.569   1.990  -2.139 0.6897520   5.771        Up
## 491 1999   5.771  -2.050   3.803  -2.569   1.990 0.8093960   0.867        Up
## 492 1999   0.867   5.771  -2.050   3.803  -2.569 0.7614250   1.105        Up
## 493 1999   1.105   0.867   5.771  -2.050   3.803 0.7420600  -4.359      Down
## 494 1999  -4.359   1.105   0.867   5.771  -2.050 0.7169820  -2.080      Down
## 495 1999  -2.080  -4.359   1.105   0.867   5.771 0.7070100  -2.140      Down
## 496 1999  -2.140  -2.080  -4.359   1.105   0.867 0.7473300   2.106        Up
## 497 1999   2.106  -2.140  -2.080  -4.359   1.105 0.7500200   0.673        Up
## 498 1999   0.673   2.106  -2.140  -2.080  -4.359 0.6606500   0.872        Up
## 499 1999   0.872   0.673   2.106  -2.140  -2.080 0.7137900   0.665        Up
## 500 1999   0.665   0.872   0.673   2.106  -2.140 0.7036200  -0.411      Down
## 501 1999  -0.411   0.665   0.872   0.673   2.106 0.7722250  -1.201      Down
## 502 1999  -1.201  -0.411   0.665   0.872   0.673 0.7561200  -4.348      Down
## 503 1999  -4.348  -1.201  -0.411   0.665   0.872 0.7942200   0.427        Up
## 504 1999   0.427  -4.348  -1.201  -0.411   0.665 0.8871600   4.148        Up
## 505 1999   4.148   0.427  -4.348  -1.201  -0.411 0.8778600  -6.632      Down
## 506 1999  -6.632   4.148   0.427  -4.348  -1.201 0.8121200   4.348        Up
## 507 1999   4.348  -6.632   4.148   0.427  -4.348 0.9249800   4.708        Up
## 508 1999   4.708   4.348  -6.632   4.148   0.427 0.9722000   0.536        Up
## 509 1999   0.536   4.708   4.348  -6.632   4.148 0.9337800   1.885        Up
## 510 1999   1.885   0.536   4.708   4.348  -6.632 0.8874600   1.858        Up
## 511 1999   1.858   1.885   0.536   4.708   4.348 0.9229000  -0.378      Down
## 512 1999  -0.378   1.858   1.885   0.536   4.708 0.7116300   1.177        Up
## 513 1999   1.177  -0.378   1.858   1.885   0.536 0.9217400  -1.134      Down
## 514 1999  -1.134   1.177  -0.378   1.858   1.885 1.0137800   0.282        Up
## 515 1999   0.282  -1.134   1.177  -0.378   1.858 1.0918800   2.626        Up
## 516 1999   2.626   0.282  -1.134   1.177  -0.378 0.8616750   0.748        Up
## 517 2000   0.748   2.626   0.282  -1.134   1.177 0.5749180  -1.891      Down
## 518 2000  -1.891   0.748   2.626   0.282  -1.134 1.0687600   1.643        Up
## 519 2000   1.643  -1.891   0.748   2.626   0.282 1.0339400  -1.624      Down
## 520 2000  -1.624   1.643  -1.891   0.748   2.626 1.1137500  -5.634      Down
## 521 2000  -5.634  -1.624   1.643  -1.891   0.748 1.1064200   4.721        Up
## 522 2000   4.721  -5.634  -1.624   1.643  -1.891 1.0410000  -2.615      Down
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## 523 2000  -2.615   4.721  -5.634  -1.624   1.643 1.0201600  -2.958      Down
## 524 2000  -2.958  -2.615   4.721  -5.634  -1.624 1.0230600  -0.946      Down
## 525 2000  -0.946  -2.958  -2.615   4.721  -5.634 1.0634750   5.686        Up
## 526 2000   5.686  -0.946  -2.958  -2.615   4.721 1.1707600  -1.001      Down
## 527 2000  -1.001   5.686  -0.946  -2.958  -2.615 1.1615800   4.975        Up
## 528 2000   4.975  -1.001   5.686  -0.946  -2.958 1.2380600   4.301        Up
## 529 2000   4.301   4.975  -1.001   5.686  -0.946 1.0384400  -1.891      Down
## 530 2000  -1.891   4.301   4.975  -1.001   5.686 1.0685600   1.186        Up
## 531 2000   1.186  -1.891   4.301   4.975  -1.001 1.1094120 -10.538      Down
## 532 2000 -10.538   1.186  -1.891   4.301   4.975 1.0625400   5.748        Up
## 533 2000   5.748 -10.538   1.186  -1.891   4.301 1.0529250   1.247        Up
## 534 2000   1.247   5.748 -10.538   1.186  -1.891 1.0070000  -1.363      Down
## 535 2000  -1.363   1.247   5.748 -10.538   1.186 0.9401400  -0.815      Down
## 536 2000  -0.815  -1.363   1.247   5.748 -10.538 0.9004800  -0.986      Down
## 537 2000  -0.986  -0.815  -1.363   1.247   5.748 0.8584400  -2.056      Down
## 538 2000  -2.056  -0.986  -0.815  -1.363   1.247 0.9196600   7.202        Up
## 539 2000   7.202  -2.056  -0.986  -0.815  -1.363 0.9818000  -1.375      Down
## 540 2000  -1.375   7.202  -2.056  -0.986  -0.815 0.8567200   0.515        Up
## 541 2000   0.515  -1.375   7.202  -2.056  -0.986 0.9803800  -1.569      Down
## 542 2000  -1.569   0.515  -1.375   7.202  -2.056 0.9666200   0.910        Up
## 543 2000   0.910  -1.569   0.515  -1.375   7.202 1.1194400   1.671        Up
## 544 2000   1.671   0.910  -1.569   0.515  -1.375 0.8375500   2.102        Up
## 545 2000   2.102   1.671   0.910  -1.569   0.515 0.9615600  -1.973      Down
## 546 2000  -1.973   2.102   1.671   0.910  -1.569 0.9513200  -4.074      Down
## 547 2000  -4.074  -1.973   2.102   1.671   0.910 1.0443800   3.031        Up
## 548 2000   3.031  -4.074  -1.973   2.102   1.671 0.9874800   0.609        Up
## 549 2000   0.609   3.031  -4.074  -1.973   2.102 0.9354600   1.351        Up
## 550 2000   1.351   0.609   3.031  -4.074  -1.973 0.8706600   0.987        Up
## 551 2000   0.987   1.351   0.609   3.031  -4.074 0.7888200   0.951        Up
## 552 2000   0.951   0.987   1.351   0.609   3.031 0.8343800  -1.727      Down
## 553 2000  -1.727   0.951   0.987   1.351   0.609 0.9450250  -1.920      Down
## 554 2000  -1.920  -1.727   0.951   0.987   1.351 1.0482400  -1.166      Down
## 555 2000  -1.166  -1.920  -1.727   0.951   0.987 1.0764600  -0.843      Down
## 556 2000  -0.843  -1.166  -1.920  -1.727   0.951 1.1334000  -1.916      Down
## 557 2000  -1.916  -0.843  -1.166  -1.920  -1.727 1.1285800  -2.471      Down
## 558 2000  -2.471  -1.916  -0.843  -1.166  -1.920 1.1521200   1.656        Up
## 559 2000   1.656  -2.471  -1.916  -0.843  -1.166 1.2167800  -1.242      Down
## 560 2000  -1.242   1.656  -2.471  -1.916  -0.843 1.1822200   3.415        Up
## 561 2000   3.415  -1.242   1.656  -2.471  -1.916 1.1850200  -4.255      Down
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## 562 2000  -4.255   3.415  -1.242   1.656  -2.471 0.9589200   0.127        Up
## 563 2000   0.127  -4.255   3.415  -1.242   1.656 1.0683200  -1.897      Down
## 564 2000  -1.897   0.127  -4.255   3.415  -1.242 0.8652425  -1.978      Down
## 565 2000  -1.978  -1.897   0.127  -4.255   3.415 0.9516260   4.156        Up
## 566 2000   4.156  -1.978  -1.897   0.127  -4.255 1.1777800  -4.215      Down
## 567 2000  -4.215   4.156  -1.978  -1.897   0.127 1.2206600  -0.473      Down
## 568 2000  -0.473  -4.215   4.156  -1.978  -1.897 1.2946800   1.097        Up
## 569 2001   1.097  -0.473  -4.215   4.156  -1.978 0.9875000  -1.661      Down
## 570 2001  -1.661   1.097  -0.473  -4.215   4.156 1.6429750   1.556        Up
## 571 2001   1.556  -1.661   1.097  -0.473  -4.215 1.2581000   1.819        Up
## 572 2001   1.819   1.556  -1.661   1.097  -0.473 1.3519000   0.924        Up
## 573 2001   0.924   1.819   1.556  -1.661   1.097 1.2123200  -0.404      Down
## 574 2001  -0.404   0.924   1.819   1.556  -1.661 1.1330800  -2.572      Down
## 575 2001  -2.572  -0.404   0.924   1.819   1.556 1.0827200  -1.006      Down
## 576 2001  -1.006  -2.572  -0.404   0.924   1.819 1.1351000  -4.277      Down
## 577 2001  -4.277  -1.006  -2.572  -0.404   0.924 1.2294750  -0.938      Down
## 578 2001  -0.938  -4.277  -1.006  -2.572  -0.404 1.2118200  -0.062      Down
## 579 2001  -0.062  -0.938  -4.277  -1.006  -2.572 1.0706400  -6.720      Down
## 580 2001  -6.720  -0.062  -0.938  -4.277  -1.006 1.3580720  -0.930      Down
## 581 2001  -0.930  -6.720  -0.062  -0.938  -4.277 1.3594500   1.799        Up
## 582 2001   1.799  -0.930  -6.720  -0.062  -0.938 1.2553800  -2.749      Down
## 583 2001  -2.749   1.799  -0.930  -6.720  -0.062 1.3402780   4.880        Up
## 584 2001   4.880  -2.749   1.799  -0.930  -6.720 1.2011750   5.026        Up
## 585 2001   5.026   4.880  -2.749   1.799  -0.930 1.3535800   0.810        Up
## 586 2001   0.810   5.026   4.880  -2.749   1.799 1.1738400   1.082        Up
## 587 2001   1.082   0.810   5.026   4.880  -2.749 1.2020600  -1.653      Down
## 588 2001  -1.653   1.082   0.810   5.026   4.880 1.0101200   3.716        Up
## 589 2001   3.716  -1.653   1.082   0.810   5.026 1.1643400  -1.089      Down
## 590 2001  -1.089   3.716  -1.653   1.082   0.810 1.0997800  -1.348      Down
## 591 2001  -1.348  -1.089   3.716  -1.653   1.082 1.1065500   0.340        Up
## 592 2001   0.340  -1.348  -1.089   3.716  -1.653 0.9662000  -4.000      Down
## 593 2001  -4.000   0.340  -1.348  -1.089   3.716 1.1897300   0.905        Up
## 594 2001   0.905  -4.000   0.340  -1.348  -1.089 1.2765240  -0.079      Down
## 595 2001  -0.079   0.905  -4.000   0.340  -1.348 1.3141520  -2.760      Down
## 596 2001  -2.760  -0.079   0.905  -4.000   0.340 0.9355025   2.107        Up
## 597 2001   2.107  -2.760  -0.079   0.905  -4.000 1.2418600  -0.397      Down
## 598 2001  -0.397   2.107  -2.760  -0.079   0.905 1.2217200  -0.415      Down
## 599 2001  -0.415  -0.397   2.107  -2.760  -0.079 1.1391000   0.707        Up
## 600 2001   0.707  -0.415  -0.397   2.107  -2.760 1.1073600  -1.992      Down
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## 601 2001  -1.992   0.707  -0.415  -0.397   2.107 1.0026800  -2.369      Down
## 602 2001  -2.369  -1.992   0.707  -0.415  -0.397 0.9795000   1.976        Up
## 603 2001   1.976  -2.369  -1.992   0.707  -0.415 1.0158600  -4.334      Down
## 604 2001  -4.334   1.976  -2.369  -1.992   0.707 0.9741000  -4.217      Down
## 605 2001  -4.217  -4.334   1.976  -2.369  -1.992 1.3367000 -11.050      Down
## 606 2001 -11.050  -4.217  -4.334   1.976  -2.369 1.9500816   7.780        Up
## 607 2001   7.780 -11.050  -4.217  -4.334   1.976 1.5956000   2.924        Up
## 608 2001   2.924   7.780 -11.050  -4.217  -4.334 1.4053600   1.892        Up
## 609 2001   1.892   2.924   7.780 -11.050  -4.217 1.3110360  -1.664      Down
## 610 2001  -1.664   1.892   2.924   7.780 -11.050 1.2490400   2.900        Up
## 611 2001   2.900  -1.664   1.892   2.924   7.780 1.2736200  -1.576      Down
## 612 2001  -1.576   2.900  -1.664   1.892   2.924 1.2390600   3.045        Up
## 613 2001   3.045  -1.576   2.900  -1.664   1.892 1.3292600   1.637        Up
## 614 2001   1.637   3.045  -1.576   2.900  -1.664 1.3194000   1.027        Up
## 615 2001   1.027   1.637   3.045  -1.576   2.900 1.0216500  -0.947      Down
## 616 2001  -0.947   1.027   1.637   3.045  -1.576 1.3121600   1.655        Up
## 617 2001   1.655  -0.947   1.027   1.637   3.045 1.4045600  -3.041      Down
## 618 2001  -3.041   1.655  -0.947   1.027   1.637 1.3707800   1.941        Up
## 619 2001   1.941  -3.041   1.655  -0.947   1.027 1.4567600   1.409        Up
## 620 2001   1.409   1.941  -3.041   1.655  -0.947 0.7561175   0.990        Up
## 621 2002   0.990   1.409   1.941  -3.041   1.655 1.2566250  -2.295      Down
## 622 2002  -2.295   0.990   1.409   1.941  -3.041 1.3060000  -1.573      Down
## 623 2002  -1.573  -2.295   0.990   1.409   1.941 1.3738400   0.506        Up
## 624 2002   0.506  -1.573  -2.295   0.990   1.409 1.4221750  -0.978      Down
## 625 2002  -0.978   0.506  -1.573  -2.295   0.990 1.5885200  -2.315      Down
## 626 2002  -2.315  -0.978   0.506  -1.573  -2.295 1.5390400   0.726        Up
## 627 2002   0.726  -2.315  -0.978   0.506  -1.573 1.2202400  -1.299      Down
## 628 2002  -1.299   0.726  -2.315  -0.978   0.506 1.3553500   3.848        Up
## 629 2002   3.848  -1.299   0.726  -2.315  -0.978 1.3838200   2.874        Up
## 630 2002   2.874   3.848  -1.299   0.726  -2.315 1.5228600   0.159        Up
## 631 2002   0.159   2.874   3.848  -1.299   0.726 1.3142600  -1.497      Down
## 632 2002  -1.497   0.159   2.874   3.848  -1.299 1.2623800  -0.114      Down
## 633 2002  -0.114  -1.497   0.159   2.874   3.848 1.1523000  -2.149      Down
## 634 2002  -2.149  -0.114  -1.497   0.159   2.874 1.1682600  -1.044      Down
## 635 2002  -1.044  -2.149  -0.114  -1.497   0.159 1.3132600   1.275        Up
## 636 2002   1.275  -1.044  -2.149  -0.114  -1.497 1.2765800  -4.342      Down
## 637 2002  -4.342   1.275  -1.044  -2.149  -0.114 1.3670200  -0.269      Down
## 638 2002  -0.269  -4.342   1.275  -1.044  -2.149 1.4086400  -1.718      Down
## 639 2002  -1.718  -0.269  -4.342   1.275  -1.044 1.2608400   4.891        Up
```

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## 640 2002   4.891  -1.718  -0.269  -4.342   1.275 1.2908600  -2.058      Down
## 641 2002  -2.058   4.891  -1.718  -0.269  -4.342 1.0809800  -1.539      Down
## 642 2002  -1.539  -2.058   4.891  -1.718  -0.269 1.1605500  -3.712      Down
## 643 2002  -3.712  -1.539  -2.058   4.891  -1.718 1.4067600  -1.972      Down
## 644 2002  -1.972  -3.712  -1.539  -2.058   4.891 1.4377640  -1.800      Down
## 645 2002  -1.800  -1.972  -3.712  -1.539  -2.058 1.3305400   0.069        Up
## 646 2002   0.069  -1.800  -1.972  -3.712  -1.539 1.8212380  -0.080      Down
## 647 2002  -0.080   0.069  -1.800  -1.972  -3.712 1.3689250  -6.839      Down
## 648 2002  -6.839  -0.080   0.069  -1.800  -1.972 1.6076160  -7.992      Down
## 649 2002  -7.992  -6.839  -0.080   0.069  -1.800 2.2750800   0.600        Up
## 650 2002   0.600  -7.992  -6.839  -0.080   0.069 2.3370880   1.337        Up
## 651 2002   1.337   0.600  -7.992  -6.839  -0.080 1.7728800   5.137        Up
## 652 2002   5.137   1.337   0.600  -7.992  -6.839 1.4743200   2.215        Up
## 653 2002   2.215   5.137   1.337   0.600  -7.992 1.3276800   1.302        Up
## 654 2002   1.302   2.215   5.137   1.337   0.600 1.2811800  -2.635      Down
## 655 2002  -2.635   1.302   2.215   5.137   1.337 1.1344400  -2.418      Down
## 656 2002  -2.418  -2.635   1.302   2.215   5.137 1.3119250  -0.460      Down
## 657 2002  -0.460  -2.418  -2.635   1.302   2.215 1.1252400  -4.992      Down
## 658 2002  -4.992  -0.460  -2.418  -2.635   1.302 1.4535600  -2.132      Down
## 659 2002  -2.132  -4.992  -0.460  -2.418  -2.635 1.5720280  -3.238      Down
## 660 2002  -3.238  -2.132  -4.992  -0.460  -2.418 1.7364200   4.339        Up
## 661 2002   4.339  -3.238  -2.132  -4.992  -0.460 1.8688640   5.874        Up
## 662 2002   5.874   4.339  -3.238  -2.132  -4.992 1.5889580   1.499        Up
## 663 2002   1.499   5.874   4.339  -3.238  -2.132 1.5262140   0.369        Up
## 664 2002   0.369   1.499   5.874   4.339  -3.238 1.4852600  -0.690      Down
## 665 2002  -0.690   0.369   1.499   5.874   4.339 1.5174800   1.687        Up
## 666 2002   1.687  -0.690   0.369   1.499   5.874 1.3745200   2.277        Up
## 667 2002   2.277   1.687  -0.690   0.369   1.499 1.6358400   0.619        Up
## 668 2002   0.619   2.277   1.687  -0.690   0.369 1.2778400  -2.572      Down
## 669 2002  -2.572   0.619   2.277   1.687  -0.690 1.4361200  -2.494      Down
## 670 2002  -2.494  -2.572   0.619   2.277   1.687 1.2957200   0.706        Up
## 671 2002   0.706  -2.494  -2.572   0.619   2.277 1.4276460  -2.273      Down
## 672 2002  -2.273   0.706  -2.494  -2.572   0.619 0.7624775   3.791        Up
## 673 2003   3.791  -2.273   0.706  -2.494  -2.572 1.1265750   2.089        Up
## 674 2003   2.089   3.791  -2.273   0.706  -2.494 1.4988800  -2.780      Down
## 675 2003  -2.780   2.089   3.791  -2.273   0.706 1.4201200  -4.478      Down
## 676 2003  -4.478  -2.780   2.089   3.791  -2.273 1.5538375  -0.662      Down
## 677 2003  -0.662  -4.478  -2.780   2.089   3.791 1.5158460  -3.040      Down
## 678 2003  -3.040  -0.662  -4.478  -2.780   2.089 1.3737200   0.627        Up
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## 679 2003   0.627  -3.040  -0.662  -4.478  -2.780 1.3399200   1.591        Up
## 680 2003   1.591   0.627  -3.040  -0.662  -4.478 1.2296750  -0.828      Down
## 681 2003  -0.828   1.591   0.627  -3.040  -0.662 1.3496800  -1.458      Down
## 682 2003  -1.458  -0.828   1.591   0.627  -3.040 1.2931800   0.528        Up
## 683 2003   0.528  -1.458  -0.828   1.591   0.627 1.5321800   7.503        Up
## 684 2003   7.503   0.528  -1.458  -0.828   1.591 1.6103460  -3.605      Down
## 685 2003  -3.605   7.503   0.528  -1.458  -0.828 1.2812000   1.778        Up
## 686 2003   1.778  -3.605   7.503   0.528  -1.458 1.4255200  -1.200      Down
## 687 2003  -1.200   1.778  -3.605   7.503   0.528 1.2880000   2.911        Up
## 688 2003   2.911  -1.200   1.778  -3.605   7.503 1.4023500   0.585        Up
## 689 2003   0.585   2.911  -1.200   1.778  -3.605 1.4802000   3.479        Up
## 690 2003   3.479   0.585   2.911  -1.200   1.778 1.5077820   0.358        Up
## 691 2003   0.358   3.479   0.585   2.911  -1.200 1.4667000   1.167        Up
## 692 2003   1.167   0.358   3.479   0.585   2.911 1.4425800  -1.173      Down
## 693 2003  -1.173   1.167   0.358   3.479   0.585 1.3976600   3.254        Up
## 694 2003   3.254  -1.173   1.167   0.358   3.479 1.6164000   2.508        Up
## 695 2003   2.508   3.254  -1.173   1.167   0.358 1.6523400   0.086        Up
## 696 2003   0.086   2.508   3.254  -1.173   1.167 1.3854200   0.716        Up
## 697 2003   0.716   0.086   2.508   3.254  -1.173 1.5085200  -1.955      Down
## 698 2003  -1.955   0.716   0.086   2.508   3.254 1.3801600   0.971        Up
## 699 2003   0.971  -1.955   0.716   0.086   2.508 1.3356500   1.262        Up
## 700 2003   1.262   0.971  -1.955   0.716   0.086 1.4582400  -0.483      Down
## 701 2003  -0.483   1.262   0.971  -1.955   0.716 1.5312200   0.540        Up
## 702 2003   0.540  -0.483   1.262   0.971  -1.955 1.4026200  -1.855      Down
## 703 2003  -1.855   0.540  -0.483   1.262   0.971 1.4456000  -0.261      Down
## 704 2003  -0.261  -1.855   0.540  -0.483   1.262 1.3274600   1.338        Up
## 705 2003   1.338  -0.261  -1.855   0.540  -0.483 1.0372940   0.241        Up
## 706 2003   0.241   1.338  -0.261  -1.855   0.540 1.2710000   1.505        Up
## 707 2003   1.505   0.241   1.338  -0.261  -1.855 1.0624200   1.327        Up
## 708 2003   1.327   1.505   0.241   1.338  -0.261 1.5163000  -0.270      Down
## 709 2003  -0.270   1.327   1.505   0.241   1.338 1.3737600   1.735        Up
## 710 2003   1.735  -0.270   1.327   1.505   0.241 1.3820220  -3.807      Down
## 711 2003  -3.807   1.735  -0.270   1.327   1.505 1.4278000   3.310        Up
## 712 2003   3.310  -3.807   1.735  -0.270   1.327 1.4726200   0.797        Up
## 713 2003   0.797   3.310  -3.807   1.735  -0.270 1.2509200   0.121        Up
## 714 2003   0.121   0.797   3.310  -3.807   1.735 1.3206400  -1.002      Down
## 715 2003  -1.002   0.121   0.797   3.310  -3.807 1.4684800   2.119        Up
## 716 2003   2.119  -1.002   0.121   0.797   3.310 1.5384400   0.238        Up
## 717 2003   0.238   2.119  -1.002   0.121   0.797 1.4184000  -0.272      Down
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## 718 2003  -0.272   0.238   2.119  -1.002   0.121 1.2989000  -1.435      Down
## 719 2003  -1.435  -0.272   0.238   2.119  -1.002 1.3310600   2.214        Up
## 720 2003   2.214  -1.435  -0.272   0.238   2.119 1.0553550   0.312        Up
## 721 2003   0.312   2.214  -1.435  -0.272   0.238 1.3857800   1.191        Up
## 722 2003   1.191   0.312   2.214  -1.435  -0.272 1.3585200   1.352        Up
## 723 2003   1.352   1.191   0.312   2.214  -1.435 1.5495200   0.664        Up
## 724 2003   0.664   1.352   1.191   0.312   2.214 0.8177825   1.149        Up
## 725 2004   1.149   0.664   1.352   1.191   0.312 1.0630250   1.207        Up
## 726 2004   1.207   1.149   0.664   1.352   1.191 1.6733400   1.602        Up
## 727 2004   1.602   1.207   1.149   0.664   1.352 1.6073600   0.151        Up
## 728 2004   0.151   1.602   1.207   1.149   0.664 1.6776750  -0.913      Down
## 729 2004  -0.913   0.151   1.602   1.207   1.149 1.7105200   1.028        Up
## 730 2004   1.028  -0.913   0.151   1.602   1.207 1.5510200   0.267        Up
## 731 2004   0.267   1.028  -0.913   0.151   1.602 1.4400400  -0.148      Down
## 732 2004  -0.148   0.267   1.028  -0.913   0.151 1.4553250   0.073        Up
## 733 2004   0.073  -0.148   0.267   1.028  -0.913 1.4418000   1.041        Up
## 734 2004   1.041   0.073  -0.148   0.267   1.028 1.3943200  -3.137      Down
## 735 2004  -3.137   1.041   0.073  -0.148   0.267 1.5361200  -0.963      Down
## 736 2004  -0.963  -3.137   1.041   0.073  -0.148 1.4836000  -0.155      Down
## 737 2004  -0.155  -0.963  -3.137   1.041   0.073 1.4458200   3.046        Up
## 738 2004   3.046  -0.155  -0.963  -3.137   1.041 1.4977000  -0.218      Down
## 739 2004  -0.218   3.046  -0.155  -0.963  -3.137 1.3675000  -0.413      Down
## 740 2004  -0.413  -0.218   3.046  -0.155  -0.963 1.4259600   0.528        Up
## 741 2004   0.528  -0.413  -0.218   3.046  -0.155 1.5328600  -2.920      Down
## 742 2004  -2.920   0.528  -0.413  -0.218   3.046 1.6315800  -0.777      Down
## 743 2004  -0.777  -2.920   0.528  -0.413  -0.218 1.5731200  -0.273      Down
## 744 2004  -0.273  -0.777  -2.920   0.528  -0.413 1.5793600  -0.195      Down
## 745 2004  -0.195  -0.273  -0.777  -2.920   0.528 1.3602600   2.480        Up
## 746 2004   2.480  -0.195  -0.273  -0.777  -2.920 1.3525400   0.162        Up
## 747 2004   0.162   2.480  -0.195  -0.273  -0.777 1.2093500   1.245        Up
## 748 2004   1.245   0.162   2.480  -0.195  -0.273 1.2098750  -0.128      Down
## 749 2004  -0.128   1.245   0.162   2.480  -0.195 1.2982000  -0.052      Down
## 750 2004  -0.052  -0.128   1.245   0.162   2.480 1.4316400  -0.798      Down
## 751 2004  -0.798  -0.052  -0.128   1.245   0.162 1.3568200  -1.117      Down
## 752 2004  -1.117  -0.798  -0.052  -0.128   1.245 1.2998250  -1.026      Down
## 753 2004  -1.026  -1.117  -0.798  -0.052  -0.128 1.3270600  -1.379      Down
## 754 2004  -1.379  -1.026  -1.117  -0.798  -0.052 1.4927000   1.429        Up
## 755 2004   1.429  -1.379  -1.026  -1.117  -0.798 1.4813600  -3.426      Down
## 756 2004  -3.426   1.429  -1.379  -1.026  -1.117 1.3803800   0.078        Up
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## 757 2004   0.078  -3.426   1.429  -1.379  -1.026 1.2644400   3.151        Up
## 758 2004   3.151   0.078  -3.426   1.429  -1.379 1.2411600   0.858        Up
## 759 2004   0.858   3.151   0.078  -3.426   1.429 1.0351200   0.529        Up
## 760 2004   0.529   0.858   3.151   0.078  -3.426 1.0331940   0.924        Up
## 761 2004   0.924   0.529   0.858   3.151   0.078 1.2733000   0.412        Up
## 762 2004   0.412   0.924   0.529   0.858   3.151 1.2593600  -1.634      Down
## 763 2004  -1.634   0.412   0.924   0.529   0.858 1.2888400   1.927        Up
## 764 2004   1.927  -1.634   0.412   0.924   0.529 1.4786400  -0.827      Down
## 765 2004  -0.827   1.927  -1.634   0.412   0.924 1.4216400  -1.242      Down
## 766 2004  -1.242  -0.827   1.927  -1.634   0.412 1.3889400  -1.124      Down
## 767 2004  -1.124  -1.242  -0.827   1.927  -1.634 1.5878200   3.145        Up
## 768 2004   3.145  -1.124  -1.242  -0.827   1.927 1.5873600   3.183        Up
## 769 2004   3.183   3.145  -1.124  -1.242  -0.827 1.6659000   1.544        Up
## 770 2004   1.544   3.183   3.145  -1.124  -1.242 1.4476800  -1.168      Down
## 771 2004  -1.168   1.544   3.183   3.145  -1.124 1.4970400   1.052        Up
## 772 2004   1.052  -1.168   1.544   3.183   3.145 1.1187950   0.720        Up
## 773 2004   0.720   1.052  -1.168   1.544   3.183 1.6092800  -0.266      Down
## 774 2004  -0.266   0.720   1.052  -1.168   1.544 1.4963800   0.522        Up
## 775 2004   0.522  -0.266   0.720   1.052  -1.168 1.7610400   1.334        Up
## 776 2004   1.334   0.522  -0.266   0.720   1.052 1.3133500   0.148        Up
## 777 2005   0.148   1.334   0.522  -0.266   0.720 0.8895200  -2.123      Down
## 778 2005  -2.123   0.148   1.334   0.522  -0.266 1.6035400  -0.141      Down
## 779 2005  -0.141  -2.123   0.148   1.334   0.522 1.4774000  -1.406      Down
## 780 2005  -1.406  -0.141  -2.123   0.148   1.334 1.6077500   0.299        Up
## 781 2005   0.299  -1.406  -0.141  -2.123   0.148 1.5966600   2.704        Up
## 782 2005   2.704   0.299  -1.406  -0.141  -2.123 1.6252280   0.189        Up
## 783 2005   0.189   2.704   0.299  -1.406  -0.141 1.4656900  -0.308      Down
## 784 2005  -0.308   0.189   2.704   0.299  -1.406 1.4877360   0.814        Up
## 785 2005   0.814  -0.308   0.189   2.704   0.299 1.5721150   0.887        Up
## 786 2005   0.887   0.814  -0.308   0.189   2.704 1.6650280  -1.803      Down
## 787 2005  -1.803   0.887   0.814  -0.308   0.189 1.5541460  -0.869      Down
## 788 2005  -0.869  -1.803   0.887   0.814  -0.308 1.7060900  -1.532      Down
## 789 2005  -1.532  -0.869  -1.803   0.887   0.814 1.9756250   0.128        Up
## 790 2005   0.128  -1.532  -0.869  -1.803   0.887 2.0899000   0.706        Up
## 791 2005   0.706   0.128  -1.532  -0.869  -1.803 1.8619840  -3.266      Down
## 792 2005  -3.266   0.706   0.128  -1.532  -0.869 2.1199760   0.831        Up
## 793 2005   0.831  -3.266   0.706   0.128  -1.532 2.1789720   0.411        Up
## 794 2005   0.411   0.831  -3.266   0.706   0.128 2.0901840   1.253        Up
## 795 2005   1.253   0.411   0.831  -3.266   0.706 2.0315680  -1.477      Down
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## 796 2005  -1.477   1.253   0.411   0.831  -3.266 1.9531060   3.053        Up
## 797 2005   3.053  -1.477   1.253   0.411   0.831 1.8836100   0.799        Up
## 798 2005   0.799   3.053  -1.477   1.253   0.411 1.6279780  -0.230      Down
## 799 2005  -0.230   0.799   3.053  -1.477   1.253 1.7730225   0.175        Up
## 800 2005   0.175  -0.230   0.799   3.053  -1.477 1.7204560   1.573        Up
## 801 2005   1.573   0.175  -0.230   0.799   3.053 1.8766700  -2.086      Down
## 802 2005  -2.086   1.573   0.175  -0.230   0.799 1.9414400   0.241        Up
## 803 2005   0.241  -2.086   1.573   0.175  -0.230 1.7967240   1.458        Up
## 804 2005   1.458   0.241  -2.086   1.573   0.175 1.8856350   1.325        Up
## 805 2005   1.325   1.458   0.241  -2.086   1.573 1.8711840   0.469        Up
## 806 2005   0.469   1.325   1.458   0.241  -2.086 1.9167100   0.041        Up
## 807 2005   0.041   0.469   1.325   1.458   0.241 1.8777680  -0.629      Down
## 808 2005  -0.629   0.041   0.469   1.325   1.458 1.9342940   0.324        Up
## 809 2005   0.324  -0.629   0.041   0.469   1.325 1.9049680  -0.868      Down
## 810 2005  -0.868   0.324  -0.629   0.041   0.469 1.7218800  -1.198      Down
## 811 2005  -1.198  -0.868   0.324  -0.629   0.041 1.6685900   1.072        Up
## 812 2005   1.072  -1.198  -0.868   0.324  -0.629 1.9502900   1.926        Up
## 813 2005   1.926   1.072  -1.198  -0.868   0.324 1.9869325  -0.288      Down
## 814 2005  -0.288   1.926   1.072  -1.198  -0.868 2.2477940  -1.827      Down
## 815 2005  -1.827  -0.288   1.926   1.072  -1.198 2.2683360   1.112        Up
## 816 2005   1.112  -1.827  -0.288   1.926   1.072 2.0758220  -2.678      Down
## 817 2005  -2.678   1.112  -1.827  -0.288   1.926 2.3807600  -0.780      Down
## 818 2005  -0.780  -2.678   1.112  -1.827  -0.288 2.3052800  -0.588      Down
## 819 2005  -0.588  -0.780  -2.678   1.112  -1.827 2.4086680   1.595        Up
## 820 2005   1.595  -0.588  -0.780  -2.678   1.112 2.3505560   1.813        Up
## 821 2005   1.813   1.595  -0.588  -0.780  -2.678 2.4881100   1.195        Up
## 822 2005   1.195   1.813   1.595  -0.588  -0.780 2.0637380   1.097        Up
## 823 2005   1.097   1.195   1.813   1.595  -0.588 2.2264120   1.601        Up
## 824 2005   1.601   1.097   1.195   1.813   1.595 1.7797775  -0.250      Down
## 825 2005  -0.250   1.601   1.097   1.195   1.813 2.2800680  -0.451      Down
## 826 2005  -0.451  -0.250   1.601   1.097   1.195 2.1210000   0.631        Up
## 827 2005   0.631  -0.451  -0.250   1.601   1.097 2.2353740   0.106        Up
## 828 2005   0.106   0.631  -0.451  -0.250   1.601 1.8889960  -1.606      Down
## 829 2006  -1.606   0.106   0.631  -0.451  -0.250 1.4472175   2.977        Up
## 830 2006   2.977  -1.606   0.106   0.631  -0.451 2.4874500   0.168        Up
## 831 2006   0.168   2.977  -1.606   0.106   0.631 2.3211120  -2.029      Down
## 832 2006  -2.029   0.168   2.977  -1.606   0.106 2.4257500   1.762        Up
## 833 2006   1.762  -2.029   0.168   2.977  -1.606 2.5924500  -1.534      Down
## 834 2006  -1.534   1.762  -2.029   0.168   2.977 2.4855920   0.234        Up
```

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## 835 2006   0.234  -1.534   1.762  -2.029   0.168 2.3375120   1.598        Up
## 836 2006   1.598   0.234  -1.534   1.762  -2.029 2.1970720   0.170        Up
## 837 2006   0.170   1.598   0.234  -1.534   1.762 2.1009800  -0.171      Down
## 838 2006  -0.171   0.170   1.598   0.234  -1.534 2.2604080  -0.451      Down
## 839 2006  -0.451  -0.171   0.170   1.598   0.234 2.2509340   2.016        Up
## 840 2006   2.016  -0.451  -0.171   0.170   1.598 2.2740800  -0.329      Down
## 841 2006  -0.329   2.016  -0.451  -0.171   0.170 2.0942040  -0.620      Down
## 842 2006  -0.620  -0.329   2.016  -0.451  -0.171 2.1706180   0.049        Up
## 843 2006   0.049  -0.620  -0.329   2.016  -0.451 2.2851820  -0.492      Down
## 844 2006  -0.492   0.049  -0.620  -0.329   2.016 1.9903100   1.719        Up
## 845 2006   1.719  -0.492   0.049  -0.620  -0.329 2.3485900  -0.051      Down
## 846 2006  -0.051   1.719  -0.492   0.049  -0.620 2.4356660   1.156        Up
## 847 2006   1.156  -0.051   1.719  -0.492   0.049 2.3923900  -2.604      Down
## 848 2006  -2.604   1.156  -0.051   1.719  -0.492 2.3353260  -1.875      Down
## 849 2006  -1.875  -2.604   1.156  -0.051   1.719 2.6483720   1.036        Up
## 850 2006   1.036  -1.875  -2.604   1.156  -0.051 2.5128080   0.630        Up
## 851 2006   0.630   1.036  -1.875  -2.604   1.156 2.3810125  -2.788      Down
## 852 2006  -2.788   0.630   1.036  -1.875  -2.604 2.6826160  -0.061      Down
## 853 2006  -0.061  -2.788   0.630   1.036  -1.875 2.7379280  -0.563      Down
## 854 2006  -0.563  -0.061  -2.788   0.630   1.036 2.2553660   2.065        Up
## 855 2006   2.065  -0.563  -0.061  -2.788   0.630 2.3676020  -0.372      Down
## 856 2006  -0.372   2.065  -0.563  -0.061  -2.788 1.8192125  -2.314      Down
## 857 2006  -2.314  -0.372   2.065  -0.563  -0.061 2.2857540   0.331        Up
## 858 2006   0.331  -2.314  -0.372   2.065  -0.563 2.4759620   3.085        Up
## 859 2006   3.085   0.331  -2.314  -0.372   2.065 2.5602980   0.063        Up
## 860 2006   0.063   3.085   0.331  -2.314  -0.372 2.5718300  -0.986      Down
## 861 2006  -0.986   0.063   3.085   0.331  -2.314 2.2930820   2.807        Up
## 862 2006   2.807  -0.986   0.063   3.085   0.331 2.2997880  -0.554      Down
## 863 2006  -0.554   2.807  -0.986   0.063   3.085 1.8319100   1.229        Up
## 864 2006   1.229  -0.554   2.807  -0.986   0.063 1.9528780  -0.922      Down
## 865 2006  -0.922   1.229  -0.554   2.807  -0.986 2.2257725   1.597        Up
## 866 2006   1.597  -0.922   1.229  -0.554   2.807 2.6888960  -0.370      Down
## 867 2006  -0.370   1.597  -0.922   1.229  -0.554 2.4098640   1.603        Up
## 868 2006   1.603  -0.370   1.597  -0.922   1.229 2.5608060   1.029        Up
## 869 2006   1.029   1.603  -0.370   1.597  -0.922 2.6394580   1.188        Up
## 870 2006   1.188   1.029   1.603  -0.370   1.597 2.3659160   0.218        Up
## 871 2006   0.218   1.188   1.029   1.603  -0.370 2.5261240   0.639        Up
## 872 2006   0.639   0.218   1.188   1.029   1.603 2.7125320  -0.947      Down
## 873 2006  -0.947   0.639   0.218   1.188   1.029 2.6921080   1.217        Up
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## 874 2006   1.217  -0.947   0.639   0.218   1.188 2.6574020   1.470        Up
## 875 2006   1.470   1.217  -0.947   0.639   0.218 2.7613560  -0.018      Down
## 876 2006  -0.018   1.470   1.217  -0.947   0.639 2.0537275  -0.303      Down
## 877 2006  -0.303  -0.018   1.470   1.217  -0.947 2.9898280   0.940        Up
## 878 2006   0.940  -0.303  -0.018   1.470   1.217 2.6861820   1.224        Up
## 879 2006   1.224   0.940  -0.303  -0.018   1.470 2.7079220  -1.144      Down
## 880 2006  -1.144   1.224   0.940  -0.303  -0.018 2.3285660   0.534        Up
## 881 2007   0.534  -1.144   1.224   0.940  -0.303 1.5411125  -0.606      Down
## 882 2007  -0.606   0.534  -1.144   1.224   0.940 3.1176733   1.491        Up
## 883 2007   1.491  -0.606   0.534  -1.144   1.224 2.8221460  -0.016      Down
## 884 2007  -0.016   1.491  -0.606   0.534  -1.144 2.7224275  -0.582      Down
## 885 2007  -0.582  -0.016   1.491  -0.606   0.534 2.7838640   1.843        Up
## 886 2007   1.843  -0.582  -0.016   1.491  -0.606 2.7795520  -0.713      Down
## 887 2007  -0.713   1.843  -0.582  -0.016   1.491 2.6869900   1.216        Up
## 888 2007   1.216  -0.713   1.843  -0.582  -0.016 2.5274980  -0.299      Down
## 889 2007  -0.299   1.216  -0.713   1.843  -0.582 2.3688900  -4.412      Down
## 890 2007  -4.412  -0.299   1.216  -0.713   1.843 3.5999640   1.130        Up
## 891 2007   1.130  -4.412  -0.299   1.216  -0.713 3.1235860  -1.133      Down
## 892 2007  -1.133   1.130  -4.412  -0.299   1.216 3.2246920   3.544        Up
## 893 2007   3.544  -1.133   1.130  -4.412  -0.299 2.9013760  -1.062      Down
## 894 2007  -1.062   3.544  -1.133   1.130  -4.412 2.8373620   1.612        Up
## 895 2007   1.612  -1.062   3.544  -1.133   1.130 2.6927975   0.630        Up
## 896 2007   0.630   1.612  -1.062   3.544  -1.133 2.6540600   2.168        Up
## 897 2007   2.168   0.630   1.612  -1.062   3.544 3.0011180   0.655        Up
## 898 2007   0.655   2.168   0.630   1.612  -1.062 2.9783940   0.773        Up
## 899 2007   0.773   0.655   2.168   0.630   1.612 3.0906940   0.015        Up
## 900 2007   0.015   0.773   0.655   2.168   0.630 2.8056760   1.122        Up
## 901 2007   1.122   0.015   0.773   0.655   2.168 2.9180380  -0.461      Down
## 902 2007  -0.461   1.122   0.015   0.773   0.655 3.0183800   1.360        Up
## 903 2007   1.360  -0.461   1.122   0.015   0.773 2.9536375  -1.866      Down
## 904 2007  -1.866   1.360  -0.461   1.122   0.015 3.0349000   1.674        Up
## 905 2007   1.674  -1.866   1.360  -0.461   1.122 2.9758140  -1.980      Down
## 906 2007  -1.980   1.674  -1.866   1.360  -0.461 3.2172320   0.053        Up
## 907 2007   0.053  -1.980   1.674  -1.866   1.360 3.2512100   1.802        Up
## 908 2007   1.802   0.053  -1.980   1.674  -1.866 2.3175625   1.441        Up
## 909 2007   1.441   1.802   0.053  -1.980   1.674 3.0666500  -1.185      Down
## 910 2007  -1.185   1.441   1.802   0.053  -1.980 3.2635400  -4.899      Down
## 911 2007  -4.899  -1.185   1.441   1.802   0.053 4.1517860  -1.775      Down
## 912 2007  -1.775  -4.899  -1.185   1.441   1.802 4.5102080   1.436        Up
```

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## 913 2007   1.436  -1.775  -4.899  -1.185   1.441 5.3423060  -0.530      Down
## 914 2007  -0.530   1.436  -1.775  -4.899  -1.185 4.3762360   2.312        Up
## 915 2007   2.312  -0.530   1.436  -1.775  -4.899 3.0536800  -0.364      Down
## 916 2007  -0.364   2.312  -0.530   1.436  -1.775 2.7245820  -1.387      Down
## 917 2007  -1.387  -0.364   2.312  -0.530   1.436 2.8522175   2.112        Up
## 918 2007   2.112  -1.387  -0.364   2.312  -0.530 2.8511180   2.796        Up
## 919 2007   2.796   2.112  -1.387  -0.364   2.312 3.3582480   0.066        Up
## 920 2007   0.066   2.796   2.112  -1.387  -0.364 3.0708000   2.020        Up
## 921 2007   2.020   0.066   2.796   2.112  -1.387 3.0117360   0.270        Up
## 922 2007   0.270   2.020   0.066   2.796   2.112 2.9434800  -3.917      Down
## 923 2007  -3.917   0.270   2.020   0.066   2.796 3.4752200   2.309        Up
## 924 2007   2.309  -3.917   0.270   2.020   0.066 3.7160660  -1.669      Down
## 925 2007  -1.669   2.309  -3.917   0.270   2.020 3.7635060  -3.706      Down
## 926 2007  -3.706  -1.669   2.309  -3.917   0.270 4.4156840   0.347        Up
## 927 2007   0.347  -3.706  -1.669   2.309  -3.917 4.0950360  -1.237      Down
## 928 2007  -1.237   0.347  -3.706  -1.669   2.309 3.6709375   2.807        Up
## 929 2007   2.807  -1.237   0.347  -3.706  -1.669 4.0964280   1.588        Up
## 930 2007   1.588   2.807  -1.237   0.347  -3.706 3.4153620  -2.440      Down
## 931 2007  -2.440   1.588   2.807  -1.237   0.347 3.7020560   1.125        Up
## 932 2007   1.125  -2.440   1.588   2.807  -1.237 3.7459000  -0.402      Down
## 933 2007  -0.402   1.125  -2.440   1.588   2.807 2.0160500  -4.522      Down
## 934 2008  -4.522  -0.402   1.125  -2.440   1.588 3.3722575  -0.752      Down
## 935 2008  -0.752  -4.522  -0.402   1.125  -2.440 4.7888020  -5.412      Down
## 936 2008  -5.412  -0.752  -4.522  -0.402   1.125 5.0064640   0.409        Up
## 937 2008   0.409  -5.412  -0.752  -4.522  -0.402 5.1009800   4.871        Up
## 938 2008   4.871   0.409  -5.412  -0.752  -4.522 4.5395420  -4.596      Down
## 939 2008  -4.596   4.871   0.409  -5.412  -0.752 4.0354580   1.405        Up
## 940 2008   1.405  -4.596   4.871   0.409  -5.412 3.7444520   0.231        Up
## 941 2008   0.231   1.405  -4.596   4.871   0.409 3.6883475  -1.661      Down
## 942 2008  -1.661   0.231   1.405  -4.596   4.871 4.0464840  -2.800      Down
## 943 2008  -2.800  -1.661   0.231   1.405  -4.596 4.4082660  -0.404      Down
## 944 2008  -0.404  -2.800  -1.661   0.231   1.405 4.8023480   3.212        Up
## 945 2008   3.212  -0.404  -2.800  -1.661   0.231 4.5919225  -1.075      Down
## 946 2008  -1.075   3.212  -0.404  -2.800  -1.661 4.0849400   4.195        Up
## 947 2008   4.195  -1.075   3.212  -0.404  -2.800 4.1755500  -2.742      Down
## 948 2008  -2.742   4.195  -1.075   3.212  -0.404 3.6633780   4.314        Up
## 949 2008   4.314  -2.742   4.195  -1.075   3.212 3.8685760   0.540        Up
## 950 2008   0.540   4.314  -2.742   4.195  -1.075 3.9397780   1.149        Up
## 951 2008   1.149   0.540   4.314  -2.742   4.195 4.0666040  -1.812      Down
```

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## 952 2008  -1.812   1.149   0.540   4.314  -2.742 3.7512440   2.670        Up
## 953 2008   2.670  -1.812   1.149   0.540   4.314 3.8095320  -3.467      Down
## 954 2008  -3.467   2.670  -1.812   1.149   0.540 3.9057240   1.777        Up
## 955 2008   1.777  -3.467   2.670  -1.812   1.149 3.8140425  -2.835      Down
## 956 2008  -2.835   1.777  -3.467   2.670  -1.812 4.3143580  -0.048      Down
## 957 2008  -0.048  -2.835   1.777  -3.467   2.670 4.5268560  -3.096      Down
## 958 2008  -3.096  -0.048  -2.835   1.777  -3.467 4.4438080  -3.001      Down
## 959 2008  -3.001  -3.096  -0.048  -2.835   1.777 5.0313200  -1.211      Down
## 960 2008  -1.211  -3.001  -3.096  -0.048  -2.835 4.8505750  -1.854      Down
## 961 2008  -1.854  -1.211  -3.001  -3.096  -0.048 5.8126320   1.710        Up
## 962 2008   1.710  -1.854  -1.211  -3.001  -3.096 6.5111240  -0.232      Down
## 963 2008  -0.232   1.710  -1.854  -1.211  -3.001 5.6634480   0.203        Up
## 964 2008   0.203  -0.232   1.710  -1.854  -1.211 5.0718900   2.857        Up
## 965 2008   2.857   0.203  -0.232   1.710  -1.854 4.1882400   0.145        Up
## 966 2008   0.145   2.857   0.203  -0.232   1.710 4.5344040  -0.462      Down
## 967 2008  -0.462   0.145   2.857   0.203  -0.232 4.0635480  -0.725      Down
## 968 2008  -0.725  -0.462   0.145   2.857   0.203 3.5300360  -3.159      Down
## 969 2008  -3.159  -0.725  -0.462   0.145   2.857 5.0175300   0.756        Up
## 970 2008   0.756  -3.159  -0.725  -0.462   0.145 6.8835840   0.270        Up
## 971 2008   0.270   0.756  -3.159  -0.725  -0.462 9.3282140  -3.331      Down
## 972 2008  -3.331   0.270   0.756  -3.159  -0.725 5.3198940  -9.399      Down
## 973 2008  -9.399  -3.331   0.270   0.756  -3.159 6.2053260 -18.195      Down
## 974 2008 -18.195  -9.399  -3.331   0.270   0.756 8.4033579   4.596        Up
## 975 2008   4.596 -18.195  -9.399  -3.331   0.270 7.3067940  -6.781      Down
## 976 2008  -6.781   4.596 -18.195  -9.399  -3.331 6.0359360  10.491        Up
## 977 2008  10.491  -6.781   4.596 -18.195  -9.399 6.4605960  -3.898      Down
## 978 2008  -3.898  10.491  -6.781   4.596 -18.195 5.2968160  -6.198      Down
## 979 2008  -6.198  -3.898  10.491  -6.781   4.596 5.8129340  -8.389      Down
## 980 2008  -8.389  -6.198  -3.898  10.491  -6.781 7.3490400  12.026        Up
## 981 2008  12.026  -8.389  -6.198  -3.898  10.491 5.8415650  -2.251      Down
## 982 2008  -2.251  12.026  -8.389  -6.198  -3.898 6.0939500   0.418        Up
## 983 2008   0.418  -2.251  12.026  -8.389  -6.198 5.9324540   0.926        Up
## 984 2008   0.926   0.418  -2.251  12.026  -8.389 5.8559720  -1.698      Down
## 985 2008  -1.698   0.926   0.418  -2.251  12.026 3.0871050   6.760        Up
test
##      Year   Lag1   Lag2   Lag3   Lag4   Lag5   Volume  Today Direction
## 986  2009  6.760 -1.698  0.926  0.418 -2.251 3.793110 -4.448      Down
## 987  2009 -4.448  6.760 -1.698  0.926  0.418 5.043904 -4.518      Down
## 988  2009 -4.518 -4.448  6.760 -1.698  0.926 5.948758 -2.137      Down
```

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## 989  2009 -2.137 -4.518 -4.448  6.760 -1.698 6.129763 -0.730      Down
## 990  2009 -0.730 -2.137 -4.518 -4.448  6.760 5.602004  5.173        Up
## 991  2009  5.173 -0.730 -2.137 -4.518 -4.448 6.217632 -4.808      Down
## 992  2009 -4.808  5.173 -0.730 -2.137 -4.518 6.008822 -6.868      Down
## 993  2009 -6.868 -4.808  5.173 -0.730 -2.137 6.401515 -4.540      Down
## 994  2009 -4.540 -6.868 -4.808  5.173 -0.730 7.550776 -7.035      Down
## 995  2009 -7.035 -4.540 -6.868 -4.808  5.173 7.592844 10.707        Up
## 996  2009 10.707 -7.035 -4.540 -6.868 -4.808 7.459436  1.585        Up
## 997  2009  1.585 10.707 -7.035 -4.540 -6.868 7.963276  6.168        Up
## 998  2009  6.168  1.585 10.707 -7.035 -4.540 6.952820  3.255        Up
## 999  2009  3.255  6.168  1.585 10.707 -7.035 6.286870  1.669        Up
## 1000 2009  1.669  3.255  6.168  1.585 10.707 6.226188  1.522        Up
## 1001 2009  1.522  1.669  3.255  6.168  1.585 6.839302 -0.388      Down
## 1002 2009 -0.388  1.522  1.669  3.255  6.168 7.083170  1.303        Up
## 1003 2009  1.303 -0.388  1.522  1.669  3.255 6.043558  5.893        Up
## 1004 2009  5.893  1.303 -0.388  1.522  1.669 7.952024 -4.988      Down
## 1005 2009 -4.988  5.893  1.303 -0.388  1.522 6.337752  0.467        Up
## 1006 2009  0.467 -4.988  5.893  1.303 -0.388 6.339728  3.623        Up
## 1007 2009  3.623  0.467 -4.988  5.893  1.303 5.788812  2.279        Up
## 1008 2009  2.279  3.623  0.467 -4.988  5.893 5.662470  0.651        Up
## 1009 2009  0.651  2.279  3.623  0.467 -4.988 4.866352 -2.640      Down
## 1010 2009 -2.640  0.651  2.279  3.623  0.467 5.114026 -0.253      Down
## 1011 2009 -0.253 -2.640  0.651  2.279  3.623 5.119916 -2.446      Down
## 1012 2009 -2.446 -0.253 -2.640  0.651  2.279 4.172433 -1.929      Down
## 1013 2009 -1.929 -2.446 -0.253 -2.640  0.651 4.673382  6.967        Up
## 1014 2009  6.967 -1.929 -2.446 -0.253 -2.640 4.785464  4.134        Up
## 1015 2009  4.134  6.967 -1.929 -2.446 -0.253 5.003300  0.839        Up
## 1016 2009  0.839  4.134  6.967 -1.929 -2.446 5.294932  2.329        Up
## 1017 2009  2.329  0.839  4.134  6.967 -1.929 6.427946 -0.632      Down
## 1018 2009 -0.632  2.329  0.839  4.134  6.967 5.373764  2.195        Up
## 1019 2009  2.195 -0.632  2.329  0.839  4.134 4.664650  0.273        Up
## 1020 2009  0.273  2.195 -0.632  2.329  0.839 5.744582 -1.218      Down
## 1021 2009 -1.218  0.273  2.195 -0.632  2.329 5.286260  2.591        Up
## 1022 2009  2.591 -1.218  0.273  2.195 -0.632 5.137923  2.452        Up
## 1023 2009  2.452  2.591 -1.218  0.273  2.195 6.046968 -2.239      Down
## 1024 2009 -2.239  2.452  2.591 -1.218  0.273 5.081302 -1.836      Down
## 1025 2009 -1.836 -2.239  2.452  2.591 -1.218 5.210080  4.514        Up
## 1026 2009  4.514 -1.836 -2.239  2.452  2.591 4.466710  1.511        Up
## 1027 2009  1.511  4.514 -1.836 -2.239  2.452 4.740370 -0.743      Down
```

```
## 1028 2009 -0.743  1.511  4.514 -1.836 -2.239 5.118466 -4.021      Down
## 1029 2009 -4.021 -0.743  1.511  4.514 -1.836 6.081714  3.195        Up
## 1030 2009  3.195 -4.021 -0.743  1.511  4.514 5.290226  2.261        Up
## 1031 2009  2.261  3.195 -4.021 -0.743  1.511 4.218872 -0.192      Down
## 1032 2009 -0.192  2.261  3.195 -4.021 -0.743 4.122504  0.010        Up
## 1033 2009  0.010 -0.192  2.261  3.195 -4.021 3.232000  1.328        Up
## 1034 2009  1.328  0.010 -0.192  2.261  3.195 4.535468  0.039        Up
## 1035 2009  0.039  1.328  0.010 -0.192  2.261 4.150876 -0.356      Down
## 1036 2009 -0.356  0.039  1.328  0.010 -0.192 5.672874  2.178        Up
## 1037 2009  2.178 -0.356  0.039  1.328  0.010 3.013263 -1.010      Down
## 1038 2010 -1.010  2.178 -0.356  0.039  1.328 2.390427  2.680        Up
## 1039 2010  2.680 -1.010  2.178 -0.356  0.039 4.223070 -0.782      Down
## 1040 2010 -0.782  2.680 -1.010  2.178 -0.356 4.363246 -3.897      Down
## 1041 2010 -3.897 -0.782  2.680 -1.010  2.178 5.654582 -1.639      Down
## 1042 2010 -1.639 -3.897 -0.782  2.680 -1.010 5.079534 -0.715      Down
## 1043 2010 -0.715 -1.639 -3.897 -0.782  2.680 5.082238  0.874        Up
## 1044 2010  0.874 -0.715 -1.639 -3.897 -0.782 4.403416  3.130        Up
## 1045 2010  3.130  0.874 -0.715 -1.639 -3.897 4.040725 -0.422      Down
## 1046 2010 -0.422  3.130  0.874 -0.715 -1.639 4.194034  3.097        Up
## 1047 2010  3.097 -0.422  3.130  0.874 -0.715 4.002330  0.991        Up
## 1048 2010  0.991  3.097 -0.422  3.130  0.874 4.805318  0.862        Up
## 1049 2010  0.862  0.991  3.097 -0.422  3.130 4.588800  0.577        Up
## 1050 2010  0.577  0.862  0.991  3.097 -0.422 4.751278  0.987        Up
## 1051 2010  0.987  0.577  0.862  0.991  3.097 4.237947  1.381        Up
## 1052 2010  1.381  0.987  0.577  0.862  0.991 4.461554 -0.188      Down
## 1053 2010 -0.188  1.381  0.987  0.577  0.862 5.974902  2.110        Up
## 1054 2010  2.110 -0.188  1.381  0.987  0.577 5.800096 -2.513      Down
## 1055 2010 -2.513  2.110 -0.188  1.381  0.987 6.310456 -6.388      Down
## 1056 2010 -6.388 -2.513  2.110 -0.188  1.381 7.683886  2.232        Up
## 1057 2010  2.232 -6.388 -2.513  2.110 -0.188 5.791750 -4.226      Down
## 1058 2010 -4.226  2.232 -6.388 -2.513  2.110 6.528052  0.158        Up
## 1059 2010  0.158 -4.226  2.232 -6.388 -2.513 5.528868 -2.252      Down
## 1060 2010 -2.252  0.158 -4.226  2.232 -6.388 5.368597  2.509        Up
## 1061 2010  2.509 -2.252  0.158 -4.226  2.232 5.369514  2.374        Up
## 1062 2010  2.374  2.509 -2.252  0.158 -4.226 4.637208 -3.646      Down
## 1063 2010 -3.646  2.374  2.509 -2.252  0.158 4.699712 -5.032      Down
## 1064 2010 -5.032 -3.646  2.374  2.509 -2.252 5.100892  5.416        Up
## 1065 2010  5.416 -5.032 -3.646  2.374  2.509 4.419372 -1.213      Down
## 1066 2010 -1.213  5.416 -5.032 -3.646  2.374 4.487664  3.548        Up
```

```
## 1067 2010  3.548 -1.213  5.416 -5.032 -3.646 4.580286 -0.096      Down
## 1068 2010 -0.096  3.548 -1.213  5.416 -5.032 4.271320  1.819        Up
## 1069 2010  1.819 -0.096  3.548 -1.213  5.416 3.963460 -3.779      Down
## 1070 2010 -3.779  1.819 -0.096  3.548 -1.213 3.906558 -0.700      Down
## 1071 2010 -0.700 -3.779  1.819 -0.096  3.548 3.777406 -0.663      Down
## 1072 2010 -0.663 -0.700 -3.779  1.819 -0.096 3.951328  3.750        Up
## 1073 2010  3.750 -0.663 -0.700 -3.779  1.819 3.718470  0.456        Up
## 1074 2010  0.456  3.750 -0.663 -0.700 -3.779 3.195238  1.446        Up
## 1075 2010  1.446  0.456  3.750 -0.663 -0.700 3.972432  2.050        Up
## 1076 2010  2.050  1.446  0.456  3.750 -0.663 3.884522 -0.212      Down
## 1077 2010 -0.212  2.050  1.446  0.456  3.750 4.037410  1.650        Up
## 1078 2010  1.650 -0.212  2.050  1.446  0.456 3.905616  0.948        Up
## 1079 2010  0.948  1.650 -0.212  2.050  1.446 4.449160  0.586        Up
## 1080 2010  0.586  0.948  1.650 -0.212  2.050 4.576282  0.015        Up
## 1081 2010  0.015  0.586  0.948  1.650 -0.212 4.116414  3.599        Up
## 1082 2010  3.599  0.015  0.586  0.948  1.650 4.798758 -2.173      Down
## 1083 2010 -2.173  3.599  0.015  0.586  0.948 4.298262  0.043        Up
## 1084 2010  0.043 -2.173  3.599  0.015  0.586 4.177436 -0.861      Down
## 1085 2010 -0.861  0.043 -2.173  3.599  0.015 3.205160  2.969        Up
## 1086 2010  2.969 -0.861  0.043 -2.173  3.599 4.242568  1.281        Up
## 1087 2010  1.281  2.969 -0.861  0.043 -2.173 4.835082  0.283        Up
## 1088 2010  0.283  1.281  2.969 -0.861  0.043 4.454044  1.034        Up
## 1089 2010  1.034  0.283  1.281  2.969 -0.861 2.707105  0.069        Up
# b) Logistic Regression Model
fit = glm(Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 + Volume,
          data = train, family = binomial)

summary(fit)
##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##     Volume, family = binomial, data = train)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.33258    0.09421   3.530 0.000415 ***
## Lag1        -0.06231    0.02935  -2.123 0.033762 *
## Lag2         0.04468    0.02982   1.499 0.134002
## Lag3        -0.01546    0.02948  -0.524 0.599933
```

```
## Lag4        -0.03111    0.02924  -1.064 0.287241
## Lag5        -0.03775    0.02924  -1.291 0.196774
## Volume      -0.08972    0.05410  -1.658 0.097240 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1354.7  on 984  degrees of freedom
## Residual deviance: 1342.3  on 978  degrees of freedom
## AIC: 1356.3
##
## Number of Fisher Scoring iterations: 4
```

We observe that, typically no predictors are stongly significant at 5% level. Usually Lag 2 test has the smallest p value.

Interpretation of coefficient corresponding to Lag 2 test :

If coefficient of Lag2 ≈ 0.01(say):

A 1 unit increase in Lag2 increases log-odds of “Up” by 0.05

Odds increase by: e^0.01≈1.010

```
# c) Odds of market going “up” at median values of the predictors.
medians = apply(train[, -9], 2, median)  # exclude Direction

log_odds = predict(fit, newdata = as.data.frame(t(medians)), type = "link")

odds = exp(log_odds)
odds
##        1
## 1.267475
```

here, Odds = P(Up)/P(Down)

```
# e) Optimal Cutoff & Test Performance.
library(pROC)
## Warning: package 'pROC' was built under R version 4.3.3
## Type 'citation("pROC")' for a citation.
##
## Attaching package: 'pROC'
## The following objects are masked from 'package:stats':
##
```

```
##     cov, smooth, var
prob_test = predict(fit, newdata = test, type = "response") # probabilities on test
data
prob_test
##       986       987       988       989       990       991       992       993
## 0.3900305 0.6084798 0.4489399 0.4101072 0.4336596 0.4406039 0.6393472 0.4987031
##       994       995       996       997       998       999      1000      1001
## 0.3946359 0.4868552 0.2988417 0.6241049 0.4061223 0.4376311 0.3244490 0.3535797
##      1002      1003      1004      1005      1006      1007      1008      1009
## 0.3610958 0.3760194 0.3112674 0.5677247 0.3531644 0.3665407 0.4427406 0.5187378
##      1010      1011      1012      1013      1014      1015      1016      1017
## 0.4754652 0.3901518 0.5084729 0.4964232 0.3842032 0.5132099 0.5086487 0.3636370
##      1018      1019      1020      1021      1022      1023      1024      1025
## 0.3987532 0.3848330 0.4511306 0.4611357 0.4030176 0.4208027 0.5282979 0.4518933
##      1026      1027      1028      1029      1030      1031      1032      1033
## 0.3610182 0.5050770 0.5147535 0.4774696 0.3259190 0.4960458 0.5447460 0.5127063
##      1034      1035      1036      1037      1038      1039      1040      1041
## 0.4146905 0.4843944 0.4585854 0.4671273 0.5578728 0.4298428 0.5177139 0.4852412
##      1042      1043      1044      1045      1046      1047      1048      1049
## 0.4431740 0.4579394 0.5068031 0.5058031 0.5481132 0.4287564 0.4636233 0.4399309
##      1050      1051      1052      1053      1054      1055      1056      1057
## 0.4534166 0.4391984 0.4544276 0.4512062 0.4009511 0.4849995 0.4625922 0.3472306
##      1058      1059      1060      1061      1062      1063      1064      1065
## 0.5516312 0.4742688 0.5585627 0.4106378 0.5175253 0.5669743 0.4989624 0.3233475
##      1066      1067      1068      1069      1070      1071      1072      1073
## 0.5864121 0.4642659 0.5384404 0.4104294 0.5586630 0.4278575 0.4983929 0.4490298
##      1074      1075      1076      1077      1078      1079      1080      1081
## 0.5889730 0.4739070 0.4556775 0.4742906 0.4442330 0.4583421 0.4579893 0.4826393
##      1082      1083      1084      1085      1086      1087      1088      1089
## 0.3959474 0.5470895 0.4451720 0.5054130 0.4157042 0.5114028 0.4879634 0.4895628
# Finding Optimal Cutoff
roc_test = roc(test$Direction, prob_test)
## Setting levels: control = Down, case = Up
## Setting direction: controls < cases
best=coords(roc_test, "best", ret = "threshold", best.method = "youden");
c_star=best[1]
# Predictions
prob_test = predict(fit, newdata = test, type = "response")


pred_class = ifelse(prob_test > c_star, "Up", "Down")
```

```
pred_class = factor(pred_class, levels = c("Down", "Up"))

# Checking lengths
length(pred_class)
## [1] 1
length(test$Direction)
## [1] 104
# Confusion matrix
str(prob_test)
##  Named num [1:104] 0.39 0.608 0.449 0.41 0.434 ...
##  - attr(*, "names")= chr [1:104] "986" "987" "988" "989" ...
str(pred_class)
##  Factor w/ 2 levels "Down","Up": 1
str(test$Direction)
##  Factor w/ 2 levels "Down","Up": 1 1 1 1 2 1 1 1 1 2 ...
mean(pred_class != test$Direction) # Test Error Rate
## [1] 0.5865385
# f) Model with Lag 1 and Lag 2 only
fit2 = glm(Direction ~ Lag1 + Lag2,
           data = train, family = binomial)

prob_test2 = predict(fit2, newdata = test, type = "response")

pred2 = ifelse(prob_test2 > 0.5, "Up", "Down")

mean(pred2 != test$Direction)
## [1] 0.4230769
```

Using only Lag1 & Lag2 gives comparable performance.

Here Test error is almost similar or slightly better.

Simple models performs well.